

**TEMPERATURE PREFERENCE, AEROBIC SCOPE AND UPPER THERMAL
TOLERANCE IN SYMPATRIC JUVENILE CHINOOK SALMON
(*ONCORHYNCHUS TSHAWYTSCHA*) AND COHO SALMON (*O. KISUTCH*)**

by

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ABSTRACT

Populations of Pacific salmon (*Oncorhynchus* spp.) are locally adapted to environmental conditions, with temperature being the most important abiotic factor. Wild Chinook salmon (*O. tshawytscha*) and coho salmon (*O. kisutch*) from the Horsefly River system in central British Columbia (BC) vary in adult migration timing, but often rear sympatrically as juveniles. These species provide an interesting opportunity to investigate adaptations to the environment in the juvenile life-history stage, which may differ from adaptations at the adult stage. I conducted three laboratory studies with wild-caught Horsefly River juvenile Chinook salmon and coho salmon to compare thermal preference, performance and tolerance between species. Temperature preference did not differ significantly between Chinook salmon and coho salmon. Aerobic scope (difference between maximum and routine metabolic rates) was also similar between the two species, and neither showed a distinct temperature optimum for peak physiological performance within the range of temperatures tested. Coho salmon had a significantly higher upper thermal limit compared to Chinook salmon, although the differences were small (0.26 °C) and may not be biologically meaningful. Thus, these two populations appear to be well adapted for their current environmental conditions rather than show any tendencies to be differentially suited to selective pressures they will experience as adults. It is important to understand how different populations and life stages of salmon will adapt and persist in warming water temperatures, knowledge which has important implications for conservation and management of salmon stocks in BC in a changing climate.

TABLE OF CONTENTS

ABSTRACT	ii
TABLE OF CONTENTS	iii
LIST OF TABLES.....	v
LIST OF FIGURES.....	vi
ACKNOWLEDGEMENTS	viii
PROLOGUE	1
CHAPTER 1.....	5
STREAM TEMPERATURES AND FISH CAPTURE	
ABSTRACT	6
INTRODUCTION	7
METHODS.....	8
Study Area	8
Temperature Loggers.....	9
Fish Capture and Transport	11
RESULTS	12
Stream Temperatures.....	12
Fish Capture.....	15
DISCUSSION.....	16
CHAPTER 2.....	19
TEMPERATURE PREFERENCE	
ABSTRACT	20
INTRODUCTION	21
METHODS.....	23
Fish Husbandry.....	23
Temperature Preference Measurements	24
Data and Statistical Analysis	26
RESULTS	28
DISCUSSION.....	31
Temperature Preference.....	31
Stream Origin and Acclimation Temperature.....	32
Ecological Relevance.....	34
CHAPTER 3.....	35
AEROBIC SCOPE	
ABSTRACT	36
INTRODUCTION	37
METHODS.....	39
Fish Husbandry	39

Metabolic Rate Measurements	39
Data and Statistical Analysis	42
RESULTS	45
DISCUSSION.....	51
Routine and Maximum Metabolic Rates	51
Aerobic Scope.....	53
Fish Size	55
Acclimation Temperature	57
Ecological Relevance.....	58
CHAPTER 4.....	61
UPPER THERMAL TOLERANCE	
ABSTRACT	62
INTRODUCTION	63
METHODS.....	65
Fish Husbandry	65
Critical Thermal Maxima Measurements	65
Data and Statistical Analysis	67
RESULTS	68
DISCUSSION.....	71
Critical Thermal Maximum	71
Stream Origin and Acclimation Temperature.....	73
Fish Size	75
Ecological Relevance.....	75
EPILOGUE	79
REFERENCES	86

LIST OF TABLES

Table 1.1 Dates of three trapping sessions which occurred in October 2016 and October 2017 in the Horsefly River watershed, with total number of traps set and trapping locations during each session.	12
Table 1.2 Average daily water temperatures of Horsefly River tributaries and mainstem for the coldest month recorded during winter 2016/2017 (Winter) and warmest month recorded during summer 2017 (Summer). *Stream orders from Holmes (2008).....	13
Table 1.3 Numbers of Chinook salmon and coho salmon trapped in each of three trapping sessions (October 21–22, 2016, October 27–28, 2016, October 21–22, 2017) in the Horsefly River watershed and numbers of each species retained and brought to the University of Northern British Columbia for laboratory experiments.	16
Table 3.1 Mean and standard deviation (SD) of routine metabolic rate (RMR), maximum metabolic rate (MMR) and absolute aerobic scope (AAS) for Chinook salmon within five temperature groups. Mean and standard deviation (SD) are presented in $\text{mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$	47
Table 3.2 Mean and standard deviation (SD) of routine metabolic rate (RMR), maximum metabolic rate (MMR) and absolute aerobic scope (AAS) for coho salmon within five temperature groups. Mean and SD are presented in $\text{mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$. Matching symbols denote significant differences ($p < 0.05$).	49

LIST OF FIGURES

Figure 1.1 Overview of the study area, the Horsefly River watershed, located approximately 75 km northeast of Williams Lake, British Columbia.....	9
Figure 1.2 Tributaries of the Horsefly River, including potential target streams: McKinley Creek, Black Creek, Wilmot Creek, Patenaude Creek, Woodjam Creek and Moffat Creek.....	10
Figure 1.3 Mean daily water temperatures in Patenaude, Wilmot and Woodjam creeks from August 6, 2016 to October 19, 2017, with error bars showing variance in daily temperature.	14
Figure 2.1 Mean temperature preference for Chinook salmon (n = 34) and coho salmon (n = 17) did not differ significantly (p = 0.09). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values, with outliers represented by open circles. Asterisks denote mean values.	28
Figure 2.2 Mean temperature preference for Chinook salmon from Patenaude Creek (n = 13) and Woodjam Creek (n = 17) did not differ significantly (p = 0.95). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values. Asterisks denote mean values.	29
Figure 2.3 Mean mass of Chinook salmon (n = 34) used for temperature preference experiments did not differ significantly from coho salmon (p = 0.32). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values, with outliers represented by open circles. Asterisks denote mean values.....	30
Figure 3.1 Metabolic responses of Chinook salmon to experimental temperatures ranging from approximately 9 °C to 21 °C. Values are (A) routine metabolic rate (n = 40) and (B) maximum metabolic rate (n = 46) measurements and calculated (C) absolute aerobic scope (n = 39) and (D) factorial aerobic scope (n = 39) for individual fish.	46
Figure 3.2 Metabolic responses of coho salmon to experimental temperatures ranging from approximately 9 °C to 21 °C. Values are (A) routine metabolic rate (n = 22) and (B) maximum metabolic rate (n = 21) measurements and calculated (C) absolute aerobic scope (n = 20) and (D) factorial aerobic scope (n = 20) for individual fish.	48
Figure 3.3 Absolute aerobic scope (mg O ₂ kg ⁻¹ h ⁻¹) values for 39 Chinook salmon and 20 coho salmon. Quadratic (Chinook) and linear (coho) regression lines were fit to aerobic scope data, with equations for best-fit lines given in the text.....	51

- Figure 4.1** Mean critical thermal maxima for coho salmon ($n = 36$) was significantly higher ($p = 0.003$) than for Chinook salmon ($n = 21$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values. Asterisks denote mean values.....69
- Figure 4.2** Critical thermal maxima of coho salmon ($n = 12$ for each) from Patenaude Creek, Wilmot Creek and Woodjam Creek did not differ significantly ($p = 0.31$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values. Asterisks denote mean values.70
- Figure 4.3** Mean mass of Chinook salmon ($n = 21$) used for upper thermal tolerance experiments was significantly heavier ($p = 0.003$) than coho salmon ($n = 36$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values, with outliers represented by open circles. Asterisks denote mean values.....71

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PROLOGUE

As ectotherms, most fishes do not regulate their internal body temperature, but body temperature varies with environmental water temperature (Fry 1947; Brett 1952, 1971). Physiological processes in fishes are predominately influenced by water temperature; therefore, it is considered the most important abiotic factor affecting fishes and has been called the “ecological master factor” (Brett 1971). Even relatively small temperature changes can have significant effects on physiological function (Angilletta et al. 2002; Newell and Quinn 2005; Farrell et al. 2008; Eliason et al. 2011; Schulte et al. 2011). Phenotypic plasticity, the capacity of a single genotype to express variable phenotypes in response to different environmental conditions, has been well-studied in fishes, including killifish (*Fundulus heteroclitus*) (e.g. Fanguie et al. 2006), minnows (*Phoxinus phoxinus*) (e.g. Killen 2014) and Pacific salmon (*Oncorhynchus* spp.) (e.g. Brett 1971).

Life history differences among species of Pacific salmon, but sympatric distributions, make species of the genus *Oncorhynchus* ideal organisms to assess responses to their environment. Pacific salmon return as adults to their natal streams to spawn; homing to these locations leads to adaptations to local environmental conditions (Dittman and Quinn 1996; Farrell et al. 2008; Eliason et al. 2011; Eliason and Farrell 2016). The capability to adjust, combined with homing behaviour, has produced numerous distinct populations of salmon throughout their geographic range (Taylor 1991; Dittman and Quinn 1996; Quinn 2005; Eliason et al. 2011; Eliason and Farrell 2016). Each population has a specific range of environmental conditions under which it functions most efficiently, tailored to maximize fitness in the local environment (Taylor 1991; Quinn 2005; Fraser et al. 2011; Eliason and

Farrell 2016). Variations in life-history strategies and adaptive capabilities among salmon species and populations in the Pacific Northwest, however, are appreciable. There is also substantial knowledge to draw on surrounding the natural history and biology of these fishes. As well, Pacific salmon are considered keystone species and exist in almost all freshwater river and lake systems in British Columbia (BC) (Hyatt and Godbout 2000). Salmon are intricately connected to ecosystems, acting as both predators and prey, and delivering critical nutrients to aquatic and riparian systems after they die (Hyatt and Godbout 2000). In addition to their intrinsic and ecological value, salmon are important for food, social and ceremonial customs in many First Nations communities in BC, as well as being an economically valuable resource in the province. Any research contributing knowledge to improve salmon conservation and management practices is therefore important.

Wild Chinook salmon (*O. tshawytscha*) and coho salmon (*O. kisutch*) from the Horsefly River system in central BC were my study populations for this thesis. Chinook salmon in the Horsefly River watershed are part of the middle Fraser River geographical grouping (DFO 1999, 2011; McPhail 2007). This population of Chinook salmon is considered secure and is not listed provincially or federally as being a conservation concern (BC Conservation Data Centre 2018a). Coho salmon from this watershed belong to a genetically distinct group known as interior Fraser coho (DFO 2002; BC MoE 2006; Holmes 2008, 2009). This population was listed by the Committee On the Status of Endangered Wildlife In Canada (COSEWIC) as endangered in 2002 (COSEWIC 2002), although recently the designation has changed to threatened after numbers increased from 2005 to 2012 (COSEWIC 2016). Interior Fraser coho salmon remain unlisted and unprotected under the federal Species At Risk Act (BC Conservation Data Centre 2018b).

Chinook salmon from the Horsefly River are of the stream-type behavioural form or race, dispersing downstream after emergence and into tributaries to rear for one to two years in fresh water prior to smoltification (Healey 1991; DFO 1999; McPhail 2007). This is in contrast to ocean-type Chinook salmon that migrate to the ocean a few months after emergence (Healey 1991; DFO 1999; McPhail 2007). Adult Chinook salmon returning to the Horsefly River migrate much earlier in the year than coho salmon, reaching their spawning grounds in the Horsefly River in early August in preparation for spawning in late August or early September (Healey 1991; DFO 1999).

Interior Fraser coho salmon emerge as fry in the spring and migrate into tributaries where they rear for approximately one year before migrating to the ocean as smolts (Sandercock 1991; DFO 2002; Irvine 2002; Hillaby 2011). In the Horsefly River watershed, coho salmon juveniles rear predominantly in narrow streams, with low water velocity and ample cover (Warren 2009). As adults, Horsefly River coho salmon enter the lower Fraser River in October and reach spawning grounds in late October to November (Sandercock 1991; DFO 2002; Irvine 2002; McPhail 2007; Hillaby 2011), migrating up river during a period of lower flow and cooler water temperatures compared to summer conditions (Sandercock 1991).

Although timing of adult migration differs between the two species, Chinook salmon and coho salmon rear as juveniles in the same tributaries within the Horsefly River watershed and are consequently subject to comparable selective pressures at this life stage (Warren 2009). The difference in adult migration timing between the two species may result in differences in selective pressures for physiological performance across a range of temperatures. Chinook salmon migrate in the summer, experiencing much warmer temperatures and higher flows compared to fall-migrating coho salmon. Based on these life-

history strategies, Horsefly River Chinook salmon and coho salmon populations provide an interesting opportunity to investigate how well they are adapted to their current environment as juveniles, versus any differences the juveniles might exhibit that may be better adapted for their lives as returning adults.

To characterize the effect of temperature on phenotype in juvenile Chinook salmon and coho salmon, I caught yearlings of both species where they rear sympatrically in tributaries to the Horsefly River in central BC. I deployed temperature loggers throughout the Horsefly River watershed in an attempt to characterize the range of water temperatures experienced by these populations during the juvenile life stage. I then conducted three laboratory studies with these wild-caught salmon to compare thermal preference, performance and tolerance between the two species. The first experiment tested temperature preference, and allowed individual fish to behaviourally choose a preferred range of water temperatures. The second experiment tested physiological performance across different experimental water temperatures with the aim of determining an optimal temperature for peak metabolic performance. The third experiment tested upper thermal tolerance of the fish to determine the maximum temperature they could withstand.

Together, these three laboratory experiments along with stream temperature records provided a better understanding of local adaptations of these two populations of salmon to their environment. Knowledge gained through the present studies on Horsefly River Chinook salmon and coho salmon is likely transferrable to other populations of salmonids in interior BC and potentially a broader range of fishes throughout the province.

CHAPTER 1

STREAM TEMPERATURES AND FISH CAPTURE

ABSTRACT

Chinook salmon (*Oncorhynchus tshawytscha*) and coho salmon (*O. kisutch*) from the Horsefly River in central British Columbia occur sympatrically during their freshwater rearing stage; therefore, they experience similar environmental conditions during this life stage. The primary objective of this study was to record water temperatures throughout the Horsefly River watershed to determine the range of temperatures that juvenile Chinook salmon and coho salmon experience. My second objective was to catch individuals from these populations for use in laboratory experiments at the University of Northern British Columbia (UNBC) to define temperature preference, performance and tolerance. To do this I deployed temperature loggers in the Horsefly River mainstem and seven of its tributaries to record water temperatures over the course of a full year. During the study period, winter water temperatures dropped to approximately 0 °C, while summer high water temperatures reached 23 °C in some locations. Smaller tributary streams were generally warmer in winter and cooler in summer compared to the mainstem and larger tributaries and therefore may serve as thermal refugia for juvenile salmon. To investigate these populations further, I trapped wild juvenile Chinook salmon and coho salmon for temperature preference, aerobic scope and upper thermal tolerance experiments at UNBC. Although my trapping effort was not equal among sites, interestingly, I caught all fishes in smaller tributaries that had the coolest summer temperatures of all sites.

INTRODUCTION

Located in central British Columbia (BC), the Horsefly River and its tributaries support populations of middle Fraser River Chinook salmon (*Oncorhynchus tshawytscha*) and interior Fraser coho salmon (*O. kisutch*) in addition to other salmonids including rainbow trout (*O. mykiss*), sockeye salmon (*O. nerka*) and kokanee (*O. nerka*), as well as numerous non-salmonids (BC MoE 2006; Holmes 2008, 2009). These populations of Chinook salmon and coho salmon migrate back to their natal streams to spawn at different times of the year, but juveniles rear in the same tributaries and are therefore exposed to similar environmental conditions during these early life stages.

With water temperature being the most important environmental factor influencing fishes (Brett 1971), it is important to characterize stream temperatures in the Horsefly River watershed, as both Chinook salmon and coho salmon remain in these rearing habitats for one to two years as juveniles before migrating to the ocean (Healey 1991; Sandercock 1991). The fry and parr life stages are critical as they account for a large portion of the life cycle and survival rates during this period are low. Published values for coho fry-to-smolt survival rates averaged on the low end 1.27 to 1.71% (reviewed in Godfrey 1965) up to approximately 5% (reviewed in Quinn and Peterson 1996). Bradford (1995) calculated egg-to-smolt survival rates for coho salmon to be similar at 1.5%. These rates were comparable to those published for Chinook salmon, which were as low as 3% for fry-to-smolt survival (reviewed in Healey 1991), and up to 6.4% for egg-to-smolt survival of stream-type Chinook salmon specifically (Bradford 1995).

The first objective of this study was to record water temperatures throughout the Horsefly River watershed for a full year to characterize the range of stream temperatures experienced by juvenile Chinook salmon and coho salmon while rearing in Horsefly River

tributaries and the mainstem river. My second objective was to catch wild Chinook salmon and coho salmon in the Horsefly River system for use in laboratory experiments at the University of Northern British Columbia (UNBC) to define temperature preference (Chapter 2), performance (Chapter 3) and tolerance (Chapter 4) in these two species.

METHODS

Study Area

Located approximately 75 km northeast of Williams Lake, BC, the Horsefly River flows westerly from its headwaters and then north into Quesnel Lake, which drains into Quesnel River and eventually meets the Fraser River (Figure 1.1). The river is approximately 98 km long, with an impassable waterfall located at kilometer 55 that acts as a migration barrier for anadromous fishes (BC MoE 2006). The Horsefly River and its tributaries provide spawning areas and juvenile rearing habitat for anadromous populations of Chinook salmon and coho salmon.

Predominant land use in the Horsefly River area, particularly within the study area downstream of the confluence of McKinley Creek, is agricultural rangeland and forestry (R.L. Case & Associates 2000; BC MoE 2006; Holmes 2008, 2009). Both industries have been operating in the Horsefly River watershed since the late 1880s, accounting for the largest anthropogenic factors influencing the watershed, with forestry perhaps playing the greatest role (Holmes 2008, 2009).

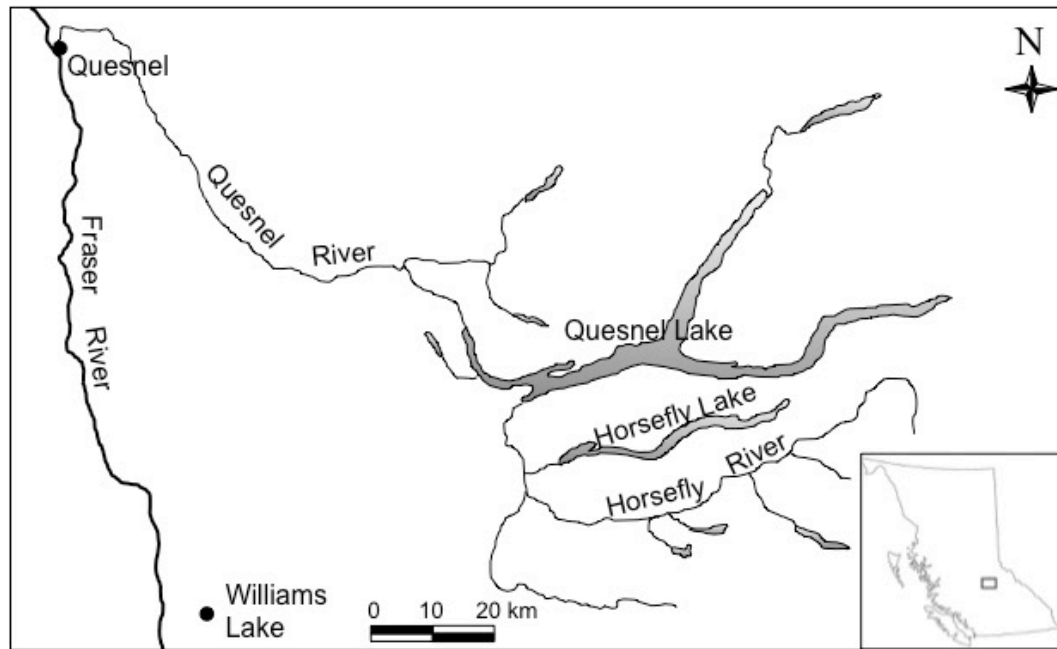


Figure 1.1 Overview of the study area, the Horsefly River watershed, located approximately 75 km northeast of Williams Lake, British Columbia.

Temperature Loggers

I identified potential target streams within the Horsefly River watershed based on results of previous fish capture in the area by other researchers (e.g. Warren 2009), as well as visual observations of fish presence while deploying temperature loggers. I initially chose seven target tributaries for wild fish capture – McKinley, Black, Wilmot, Patenaude, Woodjam, Deerhorn and Moffat creeks. In each of these streams, I deployed two HOBO U-22 temperature loggers (Onset, Bourne MA), in the area(s) identified as possible trapping locations (Figure 1.2). Temperature loggers were housed inside a 15 cm length of protective metal piping and securely attached with a cable to a fixed object on the stream bank. Loggers sat on the channel bottom; therefore, water depth of each logger varied among sites and each fluctuated with changes in seasonal water flows. I also deployed four temperature loggers in the mainstem Horsefly River in the same manner as tributary loggers. Mainstem loggers were

spaced evenly through the study area at sites with adequate access to the river. I deployed multiple loggers in each stream and the mainstem to ensure at least one logger in each location remained in the water during spring freshet, thereby providing a full year of temperature data.

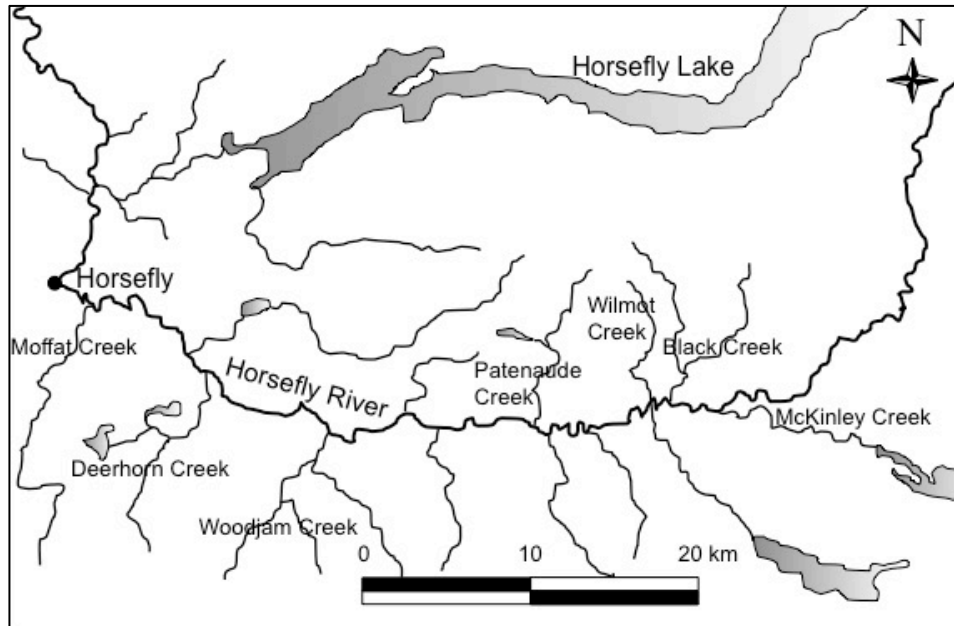


Figure 1.2 Tributaries of the Horsefly River, including potential target streams: McKinley Creek, Black Creek, Wilmot Creek, Patenaude Creek, Woodjam Creek and Moffat Creek.

Immediately prior to each fish capture session in late October 2016, I downloaded temperature loggers where I set traps to determine the range of water temperatures fishes had recently experienced. Temperature loggers remained in place to record tributary and mainstem water temperatures until October 2017.

I analyzed water temperatures based on one temperature logger from each stream as well as one logger in the Horsefly River. In many cases, only one logger in each stream remained in the water the entire year. In a few cases, both loggers in a stream recorded water

temperatures throughout the entire period, but because temperatures recorded did not vary appreciably between the two, I used data from a single, randomly-selected logger for analysis. To determine average daily water temperatures for winter 2016/2017 and summer 2017, I used the coldest month (winter) and warmest month (summer) during those periods for individual streams. As water temperatures varied among streams, the coldest and warmest months were not always defined by the same calendar months in each stream.

Fish Capture and Transport

Wild juvenile Chinook salmon and coho salmon were caught using Gee's minnow traps in two different trapping sessions in October 2016, and a single session in October 2017 (Table 1.1). I set traps in five different tributaries as well as several mainstem locations during the initial capture session, and based on catch success, locations were narrowed down to just three tributaries (Patenaude, Wilmot and Woodjam creeks) for the second and third attempts. I baited traps with cat food and in the second and third sessions, approximately half of the traps also contained salmon roe. In all three sessions, I set traps in the afternoon and retrieved them the following morning or afternoon, allowing for a total set time each session of approximately 20-23 hours.

Table 1.1 Dates of three trapping sessions, which occurred in October 2016 and October 2017 in the Horsefly River watershed, with total number of traps set and trapping locations during each session.

Date	# of Traps	Trapping Locations
October 21-22, 2016	29	McKinley, Black, Wilmot, Woodjam and Moffat creeks, Horsefly River
October 27-28, 2016	40	Patenaude, Wilmot and Woodjam creeks
October 21-22, 2017	40	Patenaude, Wilmot and Woodjam creeks

During all three trapping sessions, I quickly removed fishes from traps during retrieval, placed them into buckets with fresh river water and identified and counted each fish. I immediately released all non-target species as well as surplus coho salmon with minimal handling. I transferred juvenile Chinook salmon and coho salmon to be retained for experiments into insulated transport tanks, separated by stream. I filled transport tanks with fresh river water from where fishes were captured and equipped each tank with portable aerators. Fishes were transported by truck directly back to UNBC, with a travel time of approximately 3.5 hours, where I promptly sorted them into Recirculating Aquaculture System (RAS) tanks located in the Aquatic Animal Holding Facility upon arrival at UNBC.

RESULTS

Stream Temperatures

From summer 2016 to fall 2017, an approximate 15-month period, temperature loggers recorded water temperature once per hour in seven different tributaries as well as the Horsefly River mainstem. All tributaries and the mainstem Horsefly River dropped to temperatures close to freezing in the winter (Table 1.2). Maximum daily temperatures through the summer period averaged between 11.19 °C and 21.30 °C. Absolute maximum temperatures recorded during the summer 2017 period reached almost 24 °C (Moffat Creek).

Table 1.2 Average daily water temperatures of Horsefly River tributaries and mainstem for the coldest month recorded during winter 2016/2017 (Winter) and warmest month recorded during summer 2017 (Summer). *Stream orders from Holmes (2008).

Stream	Stream Order*	Winter – Average Daily Water Temperature (°C)			Summer – Average Daily Water Temperature (°C)		
		Minimum	Maximum	Mean	Minimum	Maximum	Mean
Black	3	0.32	0.83	0.59	9.31	16.96	12.40
Deerhorn	3	0.08	0.20	0.13	12.99	16.46	14.74
Horsefly R.	6	0.00	0.00	0.00	13.97	18.30	15.68
McKinley	5	0.35	0.75	0.53	17.96	21.30	19.36
Moffat	5	-0.04	-0.01	-0.03	13.44	20.37	16.73
Patenaude	2	0.54	1.10	0.85	7.62	11.19	9.13
Wilmot	2	0.00	0.06	0.03	9.72	15.44	12.59
Woodjam	4	-0.02	0.00	-0.01	11.75	15.71	13.64

Interestingly, the three tributaries in which I caught Chinook salmon and coho salmon, as well as those previously identified as chosen rearing habitat of juvenile coho salmon in the Horsefly River watershed (Warren 2009), had the lowest average maximum daily temperatures in summer of all streams. Absolute maximum recorded temperatures in these streams did not exceed 20 °C, reaching 13.23 °C (Patenaude Creek), 16.13 °C (Wilmot Creek) and 18.32 °C (Woodjam Creek) during the summer 2017 period (Figure 1.3).

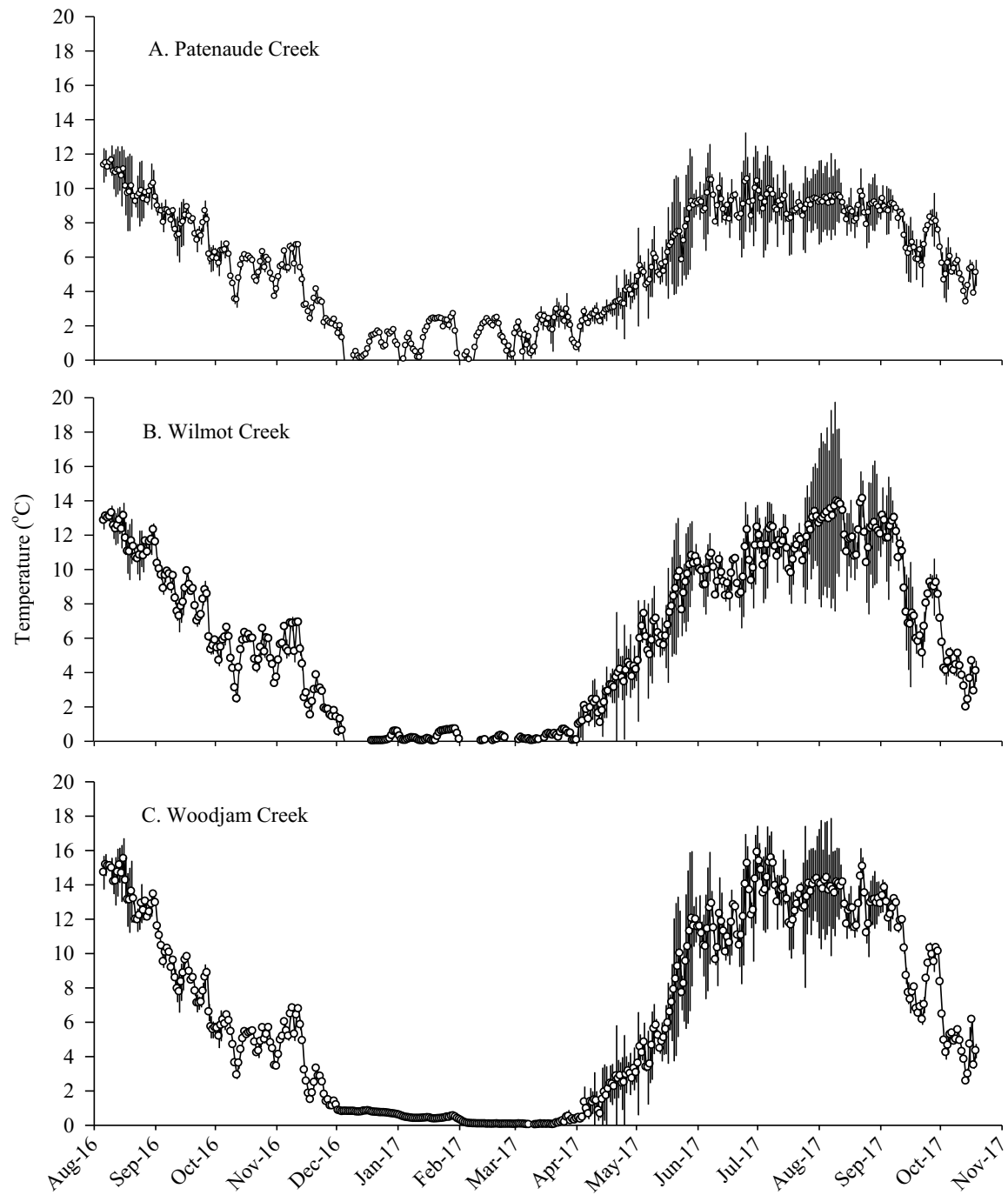


Figure 1.3 Mean daily water temperatures in Patenaude, Wilmot and Woodjam creeks from August 6, 2016 to October 19, 2017, with error bars showing variance in daily temperature.

Fish Capture

During the first fish capture session (October 21–22, 2016), I set traps in McKinley, Black, Wilmot, Woodjam and Moffat creeks, and the Horsefly River mainstem. Only six juvenile Chinook salmon were caught, all in Woodjam Creek, and all six were kept for experiments at UNBC (Table 1.3). Water temperature in Woodjam Creek at the time of fish capture was 4.6 °C. Due to low capture success, I repeated trapping again the following week (October 27–28, 2016). During the second attempt, both Chinook salmon and coho salmon were caught in each of the three tributaries trapped (Patenaude, Wilmot and Woodjam creeks), for a total of 64 Chinook salmon and 19 coho salmon. In the second session, water temperatures during fish capture in Patenaude, Wilmot and Woodjam creeks were 6.3 °C, 6.1 °C and 5.7 °C, respectively. Between the two trapping sessions in October 2016, I transported 70 Chinook salmon and 19 coho salmon to UNBC for temperature preference and aerobic scope experiments.

During the third fish capture session in October 2017, I targeted Patenaude, Wilmot and Woodjam creeks again, with much better success than the previous year. A total of 21 Chinook salmon and 457 coho salmon were caught and of those, I transported all 21 Chinook salmon and 45 coho salmon to UNBC for upper thermal tolerance experiments. Water temperatures during fish capture in 2017 in Patenaude, Wilmot and Woodjam creeks were 4.7 °C, 2.6 °C and 3.4 °C, respectively.

Non-target fishes caught during the three capture sessions were rainbow trout, redbside shiner (*Richardsonius balteatus*), northern pikeminnow (*Ptycheilus oregonensis*), and suckers (*Catostomus* spp.), all of which I immediately released.

For this project, juvenile Chinook salmon and coho salmon were only caught in Patenaude, Wilmot and Woodjam creeks in both 2016 and 2017, but this in no way infers

presence or absence of these species in other tributaries or the mainstem river as trapping effort was not equal among sites.

Table 1.3 Numbers of Chinook salmon and coho salmon trapped in each of three trapping sessions (October 21–22, 2016, October 27–28, 2016, October 21–22, 2017) in the Horsefly River watershed and numbers of each species retained and brought to the University of Northern British Columbia for laboratory experiments.

Date	Number Trapped		Number Retained	
	Chinook	Coho	Chinook	Coho
October 21–22, 2016				
Patenaude Creek	0	0	0	0
Wilmot Creek	0	0	0	0
Woodjam Creek	6	0	6	0
Total	6	0	6	0
October 27–28, 2016				
Patenaude Creek	27	3	27	3
Wilmot Creek	6	4	6	4
Woodjam Creek	31	12	31	12
Total	64	19	64	19
October 21–22, 2017				
Patenaude Creek	20	236	20	18
Wilmot Creek	0	108	0	13
Woodjam Creek	1	113	1	14
Total	21	457	21	45

DISCUSSION

Many restoration projects have been completed in the last 20 years across the Horsefly River watershed (R.L. Case & Associates 2000), which have reduced or reversed some of the damage incurred by timber harvest and the creation of agricultural land. Despite restoration efforts, it continues to be a heavily altered system that is subject to highly variable seasonal water temperatures (R.L. Case & Associates 2000; BC MoE 2006). Both industries necessitated the mass removal of trees and vegetation across huge swaths of land, including locations directly adjacent to the Horsefly River mainstem and many of its tributaries.

Removing large woody debris and vegetation from riparian zones reduces shade and cover, subjecting streams to more extreme daily and seasonal temperature fluctuations (Beschta et al. 1987; Meehan 1991; Konecki et al. 1995) and decreasing available thermal refugia for fishes (Quinn 2005; Richter and Kolmes 2005).

While there has been a history of mortality of adult salmon related to high summer water temperatures in the Horsefly River watershed (BC MoE 2006), less is known about juvenile survival through their freshwater rearing stages during peak summer temperatures. Historically, maximum summer water temperatures in the Horsefly River system have commonly reached 23 °C (R.L. Case & Associates 2000). In my study, water temperatures in several tributaries through the summer of 2017 reached approximately 23 °C as well. Patenaude, Wilmot and Woodjam creeks, the three tributaries where I caught fishes, remained slightly cooler than this in summer, with 18.3 °C being the highest temperature recorded in any of the three streams. In a previous study done in the Horsefly River watershed, Warren (2009) caught juvenile coho salmon in McKinley Creek early on in the year; however, in mid- to late-summer they migrated out of McKinley Creek and into smaller, cooler tributaries. In a study of 21 streams in northwestern California, Welsh et al. (2001) found that maximum weekly maximum temperatures of 18.0 °C was the warmest limit dictating presence of coho salmon. While there are countless ways to define and measure suitable thermal ranges for fishes, it is likely that summer temperatures in the Horsefly River system approach levels that could be potentially stressful for juvenile Chinook salmon and coho salmon that rear there.

Although I caught fishes in only three tributaries, both species are present throughout the watershed and likely move between several different locations during their freshwater rearing stage. Based on decades of fish observations in the Horsefly River watershed during

restoration projects and fish studies (reviewed in R.L. Case & Associates 2000), there are many accounts of juvenile Chinook salmon and coho salmon present in tributaries throughout the system as well as in mainstem locations (BC MoFLNRORD 2018). Shrimpton et al. (2014) examined otolith elemental signatures in juvenile Chinook salmon and coho salmon from the Horsefly and Nicola River catchments and found that both species were highly mobile, moving to an average of more than four different habitats during the freshwater life-history stage. Furthermore, individuals within each species did not show common movement patterns, suggesting that a diverse array of available habitat was used during freshwater residency (Shrimpton et al. 2014). Because of this, it was important to record water temperatures in multiple locations in the lower Horsefly River watershed to characterize temperature ranges that juvenile Chinook salmon and coho salmon may encounter during their approximately one-year-long freshwater rearing period. The Horsefly River populations of Chinook salmon and coho salmon are sympatric in their juvenile life stages, rearing in the same tributaries throughout the watershed. These fishes experience quite variable stream temperatures during the freshwater stage including potentially constraining warm summer temperatures during their first year.

CHAPTER 2

TEMPERATURE PREFERENCE

ABSTRACT

Habitat use by fishes is determined by numerous biological and ecological factors, but water temperature is thought to be the most influential determinant. Fishes use behavioural thermoregulation to exploit different thermal regimes to take advantage of optimal temperatures while avoiding unfavourable ones. In this study, I tested temperature preference in Chinook salmon (*Oncorhynchus tshawytscha*) and coho salmon (*O. kisutch*) from the Horsefly River watershed to compare this measure between species and to determine if these preferred temperatures align with important physiological requirements. To measure temperature preference, I used a shuttlebox in which two connected chambers fluctuated up and down in temperature depending on fish movement between the chambers. I calculated preferred temperature for an individual as the median temperature it resided in during the shuttling period. Preferred temperature of Chinook salmon (11.73 ± 1.70 °C) and coho salmon (12.49 ± 1.24 °C) did not differ significantly. Stream origin did not affect temperature preference in the two groups of Chinook salmon tested. Body mass of Chinook salmon and coho salmon were not significantly different and there was no relationship between fish size and temperature preference in either species. Overall, preferred temperatures of approximately 12 °C for Chinook salmon and coho salmon are consistent with previous studies done on juveniles of these species. This is also a temperature that optimizes growth under natural conditions, which is perhaps the greatest physiological priority for juvenile fishes during the freshwater rearing stage.

INTRODUCTION

The use of different habitats by fishes is influenced by a combination of biological and ecological factors that covary (Brett 1971; Konecki et al. 1995; Sauter et al. 2001; Angilletta et al. 2002). Water temperature, arguably the most important abiotic factor influencing fishes (Brett 1971), is perhaps the largest determinant of habitat use (Creaser 1930; Stauffer et al. 1976; Magnuson et al. 1979). Behavioural thermoregulation allows fishes to take advantage of optimal temperatures while avoiding unfavourable ones (Neill et al. 1972; Moyle and Cech 2004; Newell and Quinn 2005), thereby permitting them to occupy specific thermal niches that ultimately contribute to an individual's fitness (Magnuson et al. 1979).

Fry (1947) was the first to define “temperature preference” or “preferred temperature” as the terms are used today – a temperature region, within an infinite gradient, that individuals of a population will congregate with adequate precision. Around this time, researchers first began experimentally determining temperature preference in laboratory settings with both non-salmonids (Doudoroff 1938; Ferguson 1958), and salmonids (Brett 1952). Preferred temperature is thought to align closely with temperatures that optimize physiological functions such as growth or physical activity requirements (Brett 1971; Magnuson et al. 1979; Jobling 1981; Coutant 1987). These physiological requirements dictate temperature preference both daily and across life-history stages.

Fishes occupy a range of temperatures daily, and diel migrations among areas of varying water temperature may be associated with food acquisition and digestion (Brett 1971; Armstrong et al. 2013; Goyer et al. 2014). Diel migrations are most pronounced in lake and ocean environments, where fishes migrate vertically at different times of the day, moving to areas of higher food abundance to feed, and then to different areas to maximize metabolic

efficiency (Quinn 2005). Although horizontal migrations are less well-studied, Armstrong et al. (2013) showed juvenile coho salmon used long-distance horizontal movements within a stream to take advantage of thermal heterogeneity – occupying colder habitats to feed, then moving to warmer areas of the stream to increase metabolism.

Temperature preference will also dictate seasonal movements in stream habitats. During summer, elevated water temperatures force salmonids to seek out cold-water refugia, which are pockets of cooler water formed by inflowing tributaries or groundwater upwelling (Breau et al. 2011; Kurylyk et al. 2015). In winter, fishes will move into areas free of ice and with sufficient dissolved oxygen, but perhaps the largest driver is choosing water temperatures that minimize energy expenditures (Cunjak 1996).

In addition to occupying favourable temperatures on a daily and seasonal time scale, temperature preference in fishes has also been shown to vary with age, as behavioural thermoregulation reflects the importance of different priorities during particular life stages (Coutant 1987; Sauter et al. 2001; Newell and Quinn 2005; Morita et al. 2010b, 2010a; Goyer et al. 2014). Somatic growth is a driving factor for juvenile fishes as well as adults in the marine phase of their life cycles, and fishes will regulate body temperature accordingly by exploiting different thermal regimes (Morita et al. 2010a, 2010b). Younger salmonids have been shown to prefer warmer temperatures than older fishes which choose to reside in cooler water (Morita et al. 2010a). This pattern can be largely attributed to age-specific growth rates as fishes are likely to prefer temperatures that optimize growth at each life-history stage (Konecki et al. 1995b). Younger, smaller fishes grow more efficiently in warmer temperatures compared to older and larger fish which maximize growth potential in cooler environments (Morita et al. 2010b).

In contrast, spawning adults cease feeding and energy is directed toward gonadal maturation and the extreme physiological demands of migration to spawning grounds, while somatic growth is discontinued (Newell and Quinn 2005; Mathes et al. 2009). Spawning adults will therefore choose very different thermal regimes than younger fishes, and will time migrations to take advantage of preferable water temperatures that will optimize spawning potential as well as maximize incubation survival of eggs and success of emerging juveniles (Newell and Quinn 2005; Mathes et al. 2009). Thus, fishes at different life stages may have very different temperature preferences for a range of physiological and behavioural reasons.

The objective of this study was to define the behavioural temperature preference of Horsefly River juvenile Chinook salmon (*Oncorhynchus tshawytscha*) and coho salmon (*O. kisutch*). I also wanted to compare this measure between the two study species because they rear sympatrically as juveniles and therefore experience the same stream temperatures during this life stage.

METHODS

Fish Husbandry

Chinook salmon and coho salmon were caught in October 2016, as described in Chapter 1, and held in the Aquatic Animal Holding Facility (AAHF) at the University of Northern British Columbia (UNBC) for the duration of laboratory experiments. The Animal Care and Use Committee at UNBC approved fish holding conditions and experimental procedures (protocol #2016-12). Fishes caught in October 2016 were held for approximately five weeks following capture prior to experiments commencing.

The AAHF was equipped with two Recirculating Aquaculture System (RAS) units, consisting of either six or nine tanks each. I separated fishes into tanks by stream origin, but not by species. Both RAS systems had heating and cooling capabilities and five stages of filtration. Each RAS unit was equipped with sensors to measure pH, conductivity and water temperature. I programmed lights in the AAHF room for a 12-h light and 12-h dark regime. I performed daily water exchanges of at least 10%, but usually closer to 20%, for each system when in use.

Fishes were held at a water temperature of 9 ± 1 °C while awaiting or recovering from experiments. In October 2016, I initially fed fishes both frozen brine shrimp and tubifex worms, alternating between them daily. Fishes ate tubifex worms more readily and uneaten worms were much easier to siphon up than brine shrimp; therefore, I fed fishes exclusively worms after the initial two weeks of holding. I monitored and adjusted feeding to ensure all food was eaten but no excess food accumulated in tanks. Once experiments commenced, I fed fishes at 0.75-1% of their body mass once per day, adjusting ration amounts based on visual observations of fish growth and behaviour. Fishes were fasted for 24 h prior to temperature preference experiments.

Temperature Preference Measurements

I conducted temperature preference experiments between December 16th, 2016 and April 4th, 2017 and used approximately half of the Chinook salmon (36) and all of the coho salmon (19) caught in October 2016. I rotated through stream origin during experiments to standardize holding duration among the three stream populations. Within each stream, I selected individual fish randomly for temperature preference experiments.

Temperature preference was determined using a “shuttlebox” (Loligo Systems, Viborg, Denmark). The shuttlebox was composed of two adjoining 40 cm diameter circular chambers, connected by a narrow passage. For this experiment, the two chambers remained 1 °C different in temperature throughout the experiment during both “static” and “dynamic” modes. An overhead uEYE camera connected to a laptop running ShuttleSoft (v.2.6.4) software (Loligo Systems) recorded fish position and velocity once every second. I mounted two temperature probes in the shuttlebox, one within each circular chamber. The temperature probes were connected to their own temperature analyzer box, associated with either the “increasing” (warmer) chamber or the “decreasing” (cooler) chamber. Temperature analyzers were connected to a data acquisition system (DAQ-S, Loligo Systems), which communicated with the ShuttleSoft software. The DAQ-S controlled four different pumps that fed water through cold or hot baths to buffer tanks that supplied each shuttlebox chamber. Two additional pumps recirculated water from the two chambers back to the buffer tanks.

I tested fish individually in the shuttlebox, half of which I first introduced into the increasing chamber and the other half into the decreasing chamber, alternating between experiments. After a three-hour acclimation period on static mode, where temperatures in the two tanks remained stable, I switched the shuttlebox to dynamic mode, where ShuttleSoft triggered heating or chilling of the tanks in response to fish movement. When the fish was in the warmer chamber, both chambers warmed at a rate of 15 °C per hour. When the fish entered the cooler tank, both sides cooled by 15 °C per hour. The total time for each experiment lasted approximately 22 hours, ending the following morning. I then gradually brought the fish back to an ambient temperature of 9 °C, removed it from the shuttlebox, weighed and measured it, and then returned it to a holding tank.

Between each trial, I drained the shuttlebox completely and re-filled it with fresh dechlorinated water. Every one to two weeks, I sterilized the entire system with Ovadine solution (Syndel, Canada).

Data and Statistical Analysis

Out of all fish tested, two Chinook salmon and two coho salmon failed to shuttle back and forth during dynamic mode in the shuttlebox. In all four cases, the fish remained in the cold tank for most or all of the experimental period and moved very little during this time. These four trials, therefore, were excluded from further analysis, and final sample sizes for determination of temperature preference were 34 Chinook salmon and 17 coho salmon.

For each fish, I calculated temperature preference as the median temperature in which the fish resided during dynamic mode from 5 pm to 9:30 am (a total of 16.5 hours). I excluded the first two hours of dynamic mode (3 pm to 5 pm) from calculations as the fish adjusted to newly fluctuating temperatures in the chambers. The majority of fish shuttled well the entire time, and I included all 16.5 h in calculations of median temperature. Occasionally, a fish entered the colder chamber and remained there for up to several hours with the water temperature holding at approximately 4 °C, the minimum temperature limit set in ShuttleSoft. During these periods, the fish would be almost motionless as its metabolism slowed while failing to shuttle back to the warmer chamber. During successful shuttling, fish would move back and forth between chambers many times per minute. Because of this, I decided that these extended periods “stuck” in the colder tank were not representative of choosing a preferred temperature range; therefore, if a fish remained in the colder chamber and failed to shuttle for at least 1.5 h, I excluded that period from calculations. In these cases,

the length of successful shuttling used to determine temperature preference was always greater than seven hours. I calculated final temperature preferences for Chinook salmon and coho salmon as the mean of individual temperature preferences within each species.

In addition to comparing temperature preference between species, I also wanted to investigate if temperature preference differed with stream origin. The only two populations with large enough sample sizes for this comparison were Chinook salmon from Patenaude Creek and Woodjam Creek.

Sizes of individual fish used for temperature preference experiments varied considerably, therefore I compared fish mass between the two species. I also investigated whether or not temperature preference and fish mass were correlated within either of the species.

I used R version 3.5.0 (R Core Team 2018) for all statistical analyses and determined statistical significance using an alpha value of 0.05. Means are presented $\pm$ 1 standard deviation (SD). I used Shapiro-Wilk normality tests for all assessments of data normality. To test for equality of variances, I used an F-test when both data sets were normally distributed, and a Fligner-Killeen test of homogeneity of variances when one or both data sets were not normally distributed. In the comparison of two groups, I used an independent samples t-test when both data sets were normally distributed, and the non-parametric Wilcoxon Rank Sum (also known as Mann-Whitney U) test when either one or both data sets were not normally distributed, or outliers were present that could not be excluded as erroneous measurements. Finally, to test for relationships between fish mass and preferred temperature in each species, I used a Spearman's Rank-Order Correlation.

RESULTS

Temperature preferences for individual Chinook salmon ($n = 34$) varied between 8.26 °C and 15.66 °C, with a mean temperature preference of 11.73 ± 1.70 °C for the group.

Temperature preferences for 17 coho salmon individuals ranged from 11.08 °C to 15.91 °C, with a mean of 12.49 ± 1.24 °C for the group. Results of a Wilcoxon Rank Sum test indicated that there was no significant difference between mean preferred temperature of Chinook salmon and coho salmon ($W_{49} = 374$, $p = 0.09$; Figure 2.1).

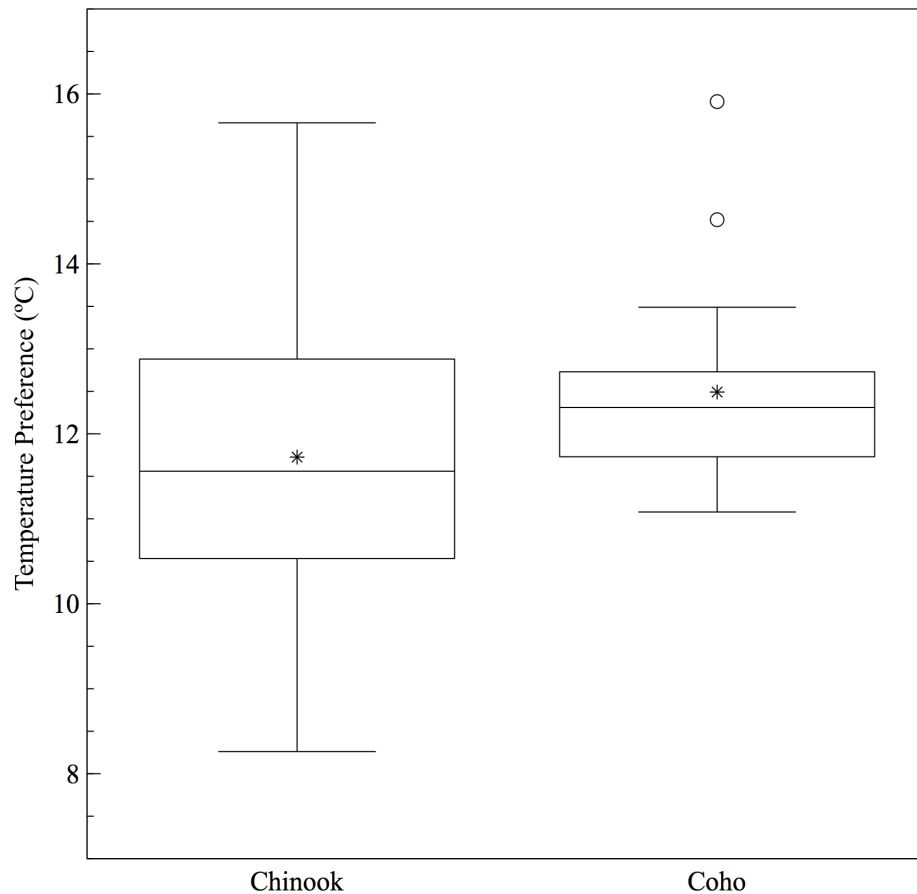


Figure 2.1 Mean temperature preference for Chinook salmon ($n = 34$) and coho salmon ($n = 17$) did not differ significantly ($p = 0.09$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values, with outliers represented by open circles. Asterisks denote mean values.

Mean temperature preference of Chinook salmon from Patenaude Creek ($n = 13$) was 11.51 ± 1.76 °C, and from Woodjam Creek ($n = 17$) was 11.47 ± 1.45 °C. An independent samples t-test detected no significant difference in temperature preference of Chinook salmon from Patenaude Creek compared to Chinook salmon from Woodjam Creek ($t_{28} = 0.07$, $p = 0.95$; Figure 2.2).

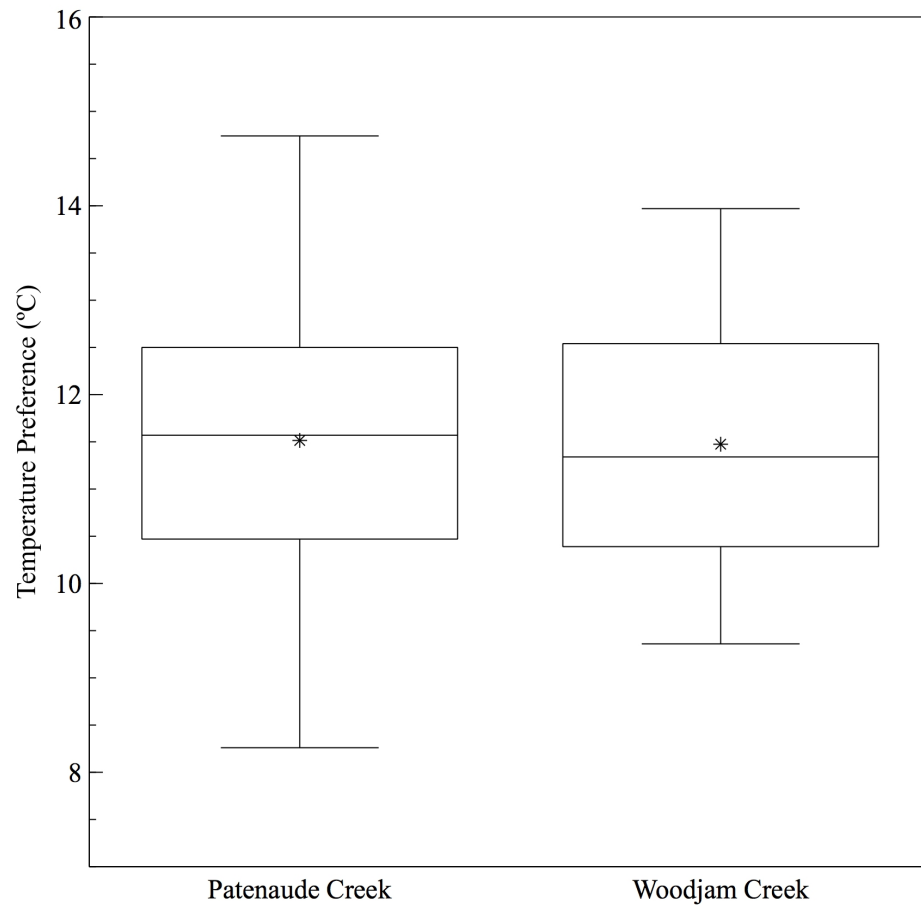


Figure 2.2 Mean temperature preference for Chinook salmon from Patenaude Creek ($n = 13$) and Woodjam Creek ($n = 17$) did not differ significantly ($p = 0.95$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values. Asterisks denote mean values.

The 34 Chinook salmon used for temperature preference experiments weighed an average of 7.70 ± 2.69 g, while coho salmon ($n = 17$) had a mean mass of 8.35 ± 2.89 g at the

time of experiments. Results of a Wilcoxon Rank Sum test indicated that there was no significant difference between mass of Chinook salmon and mass of coho salmon used for temperature preference experiments ($W_{49} = 339$, $p = 0.32$; Figure 2.3). In investigating possible relationships between temperature preference and size of the fish within each species, Spearman's Rank-Order correlations determined that there was no significant correlation between preferred temperature and body mass in Chinook salmon ($r_s(32) = -0.08$, $p = 0.64$), or coho salmon ($r_s(15) = 0.16$, $p = 0.53$).

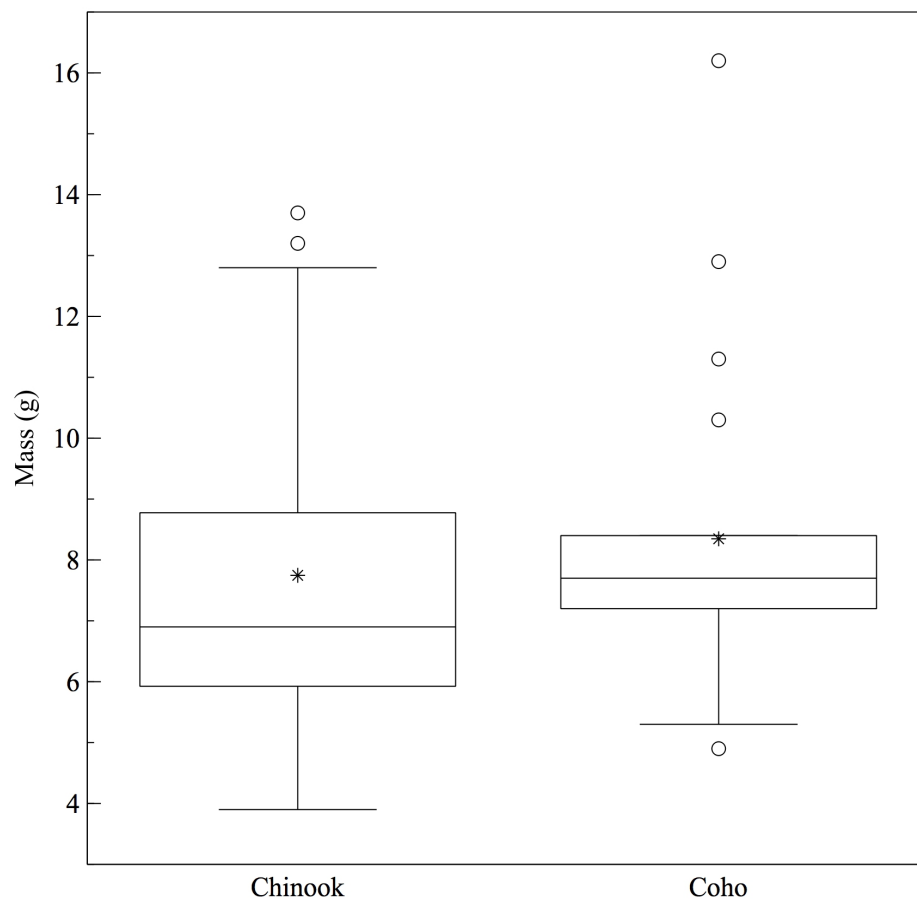


Figure 2.3 Mean mass of Chinook salmon ($n = 34$) used for temperature preference experiments did not differ significantly from coho salmon ($p = 0.32$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values, with outliers represented by open circles. Asterisks denote mean values.

DISCUSSION

Temperature Preference

Temperature preference of Chinook salmon (mean = 11.73 °C) and coho salmon (mean = 12.49 °C) determined in these experiments did not differ significantly from each other and were consistent with values in the literature for the parr life stage. Brett (1952) determined temperature preference in sub-yearling Chinook salmon to be between 12 and 13 °C with a mean temperature preference of 11.7 °C. Juvenile coho salmon have been shown to prefer a similar temperature range of 12 to 14 °C (Brett 1952). Konecki et al. (1995b) determined temperature preferences of between 10 and 12 °C for two populations of juvenile coho salmon from coastal streams in Washington. All five species of Pacific salmon have been found to behaviourally avoid temperatures above 15 °C (Brett 1952; Konecki et al. 1995; Richter and Kolmes 2005). While most Chinook salmon and coho salmon in my experiments did experience temperatures much higher than 15 °C (some as high as 20 °C) during shuttlebox trials, it was usually only for a brief time as fish would then move into the cooler tank and temperatures would begin to decrease. Only one Chinook salmon and one coho salmon had median preferred temperatures of between 15 and 16 °C, with all other individuals preferring less than 15 °C.

These results are not surprising when one considers the physiological requirements for juvenile salmon during the freshwater rearing stage. Many studies have shown optimal growth of salmon during this life-history stage to be approximately 15 °C when fed to satiation under laboratory conditions (e.g. Brett et al. 1969; Jobling 1981). In natural stream environments, however, food is usually limited and the optimal temperature for growth decreases to balance the higher metabolic demand at warmer temperatures (Brett et al. 1969;

Beschta et al. 1987). Under these conditions, a more reasonable estimate of optimal growth temperature is several degrees cooler (Brett et al. 1969; Reiser and Bjornn 1979; Beschta et al. 1987; Bell 1990; Richter and Kolmes 2005), which coincides with temperature preference of Horsefly River Chinook salmon and coho salmon found in this study, as well as in previous studies for juvenile Pacific salmon (Brett 1952).

Stream Origin and Acclimation Temperature

During the freshwater rearing stage, both Chinook salmon and coho salmon juveniles are quite mobile, moving into several different tributaries within the Horsefly River watershed (R.L. Case & Associates 2000; Shrimpton et al. 2014; BC MoFLNRORD 2018). Individuals of each species found in different streams are genetically similar, but stream origin and different environmental conditions may influence temperature preference. To investigate any differences in temperature preference among streams, only Chinook salmon from Patenaude Creek (n = 13) and Woodjam Creek (n = 17) provided large enough sample sizes for the comparison. Chinook salmon from these two creeks did not differ significantly in their preferred temperature. This result was not surprising, as mean daily water temperatures in the three streams where fishes were trapped were similar (approximately 5.0–5.5 °C) over the two-week period prior to capture. Stream temperatures in Patenaude, Wilmot and Woodjam creeks were also quite similar over the two-month period prior to fish capture, ranging from 7.0–7.4 °C. In addition to similar thermal histories experienced within the streams, all fishes were held under identical conditions for several weeks in the AAHF before experiments commenced.

Although there were negligible differences in thermal history between species and among streams in this study, acclimation temperature is an important consideration especially when conducting laboratory studies investigating temperature effects. Fry (1947) suggested that prior thermal history may influence preferred temperature, and coined a second term, “final preferendum,” which is the ultimate congregation temperature for the population based on a measure of central tendency for the group and irrespective of acclimation temperature prior to experiments. While some studies have found acclimation temperature to affect temperature preference in various species of fish (Ferguson 1958; Cherry et al. 1977), a number of studies have found no effect. In laboratory experiments with juvenile salmon, Brett (1952) found little difference in temperature preference across a 15 °C range of acclimation temperatures. Konecki et al. (1995b) tested temperature preference in two different populations of coho salmon and reported a slight difference in temperature preference with stream origin, although the difference was not significant. Coho salmon from a colder and less variable stream had a mean preferred temperature of 9.9 °C, while coho salmon from a warmer and more variable stream preferred 11.6 °C (Konecki et al. 1995b). The discrepancy in whether or not acclimation temperature and thermal history influences temperature preference can possibly be attributed to the different experimental methods used, as suggested by Reynolds and Casterlin (1979). They found that temperature preferences in acute trials, where fishes were exposed to a temperature gradient for less than two hours, would likely vary with acclimation temperature. In trials that lasted longer, approximately 24 hours or up to a few days, individuals gravitated to species-specific regions of preferred temperature regardless of prior thermal history or acclimation (Reynolds and Casterlin 1979).

Ecological Relevance

Preferred temperature in fishes is well studied in laboratory settings and is a good indicator of temperature regimes that those fishes would choose in the wild (Ferguson 1958; Neill et al. 1972; Jobling 1981). In general, fishes will behaviourally choose temperatures at which they perform optimally (Brett 1971; Magnuson et al. 1979; Jobling 1981; Coutant 1987). Temperature preference will shift at different life-history stages to reflect changing physiological demands and priorities (Coutant 1987; Sauter et al. 2001; Newell and Quinn 2005; Morita et al. 2010a, 2010b; Goyer et al. 2014). Juvenile Chinook salmon and coho salmon in this study preferred temperatures that are optimal for growth under natural conditions during this life stage. Another important measure of physiological performance is aerobic scope, which is the difference between maximum and routine metabolic rates and indicates scope for energetically expensive activities above required routine needs (Farrell 2002, 2008; Cooke et al. 2012; Rosewarne et al. 2016). I tested aerobic scope in these two populations (Chapter 3) to see if there was an optimal temperature where aerobic scope was maximized and if it aligned with these temperature preference results.

CHAPTER 3

AEROBIC SCOPE

ABSTRACT

Aerobic scope can be considered as the capacity available to complete energetically expensive activities, such as foraging and digestion, and is commonly measured in fishes to assess physiological performance. The objectives of this study were to determine how aerobic scope changes with temperature in juvenile Chinook salmon (*Oncorhynchus tshawytscha*) and coho salmon (*O. kisutch*) from the Horsefly River, and whether there is an optimal temperature at which aerobic scope is maximized in each species. I performed respirometry experiments to measure maximum (MMR) and routine metabolic rates (RMR) of individual fish. From these values I estimated two measures of aerobic scope, absolute aerobic scope ($AAS = MMR - RMR$) and factorial aerobic scope ($FAS = MMR/RMR$). In both species, AAS was either maintained or continued to increase with temperature across the range of experimental temperatures (9–21 °C) with no obvious peak present. There was no pattern in FAS across the range of temperatures in either species, but all values except for one Chinook salmon were above 2, which is generally thought to be the lower limit for aerobic capacity. Among five experimental temperature groups there were no significant differences in AAS of Chinook salmon, although mean AAS in coho salmon at 15 °C was significantly lower than at 21 °C. Controversy exists surrounding the ecological relevance of aerobic scope measurements and how experimentally determined values in a laboratory setting might translate into fish performance in a natural setting. While many competing factors exist that dictate which habitats a fish will exploit in the wild, the measure still holds value in predicting physiological performance with respect to temperature.

INTRODUCTION

During intense physical activity, salmon increase oxygen uptake and cardiac output to supply locomotory muscles with additional oxygen (Farrell et al. 2008; Cooke et al. 2012; Norin et al. 2014). A common physiological measure used to gauge performance is *aerobic scope*, which is the maximum oxygen delivered above that required for routine needs, supporting energetically expensive activities such as digestion, locomotion, growth and reproduction (Farrell 2002; Farrell et al. 2008; Cooke et al. 2012; Rosewarne et al. 2016).

Like most physiological processes in poikilotherms, aerobic scope is directly affected by temperature. An idea first pioneered by Fry (1947), aerobic scope is thought to increase until reaching a peak at an optimal temperature for performance (T_{opt}) and then steadily decrease until reaching an often-fatal critical temperature (T_{crit}). The oxygen- and capacity-limited thermal tolerance (OCLTT) hypothesis provides a mechanistic framework for explaining the shape of the “Fry curve” as aerobic scope responds to temperature (Pörtner 2002; Pörtner and Knust 2007; Pörtner and Farrell 2008). The OCLTT hypothesis attributes increasing aerobic scope to an increase in heart rate, which drives oxygen transport; however, cardiac collapse will begin at temperatures warmer than T_{opt} causing a decline in aerobic scope until it reaches zero at T_{crit} (Pörtner and Farrell 2008; Eliason et al. 2011; Eliason et al. 2013; Anttila et al. 2013; Farrell 2013; Muñoz et al. 2014).

The effect of temperature on aerobic scope follows a common pattern known as a thermal performance curve (TPC), which visually depicts how biological processes vary across a thermal range (Schulte et al. 2011). An aerobic scope TPC identifies the T_{opt} at which aerobic scope is maximized as well as the range of temperatures that support performance above a certain threshold, often arbitrarily set at 80% (Schulte et al. 2011) or

90% (Anttila et al. 2013) of maximum performance at T_{opt} . This range defines a zone of thermal tolerance.

Adult salmon from populations with more challenging migratory conditions (i.e., greater distance, higher flow, warmer water temperature) have been shown to possess greater aerobic scopes, which are maximized at warmer temperatures (Lee et al. 2003; Farrell 2007; Farrell et al. 2008; Eliason et al. 2011). Salmon populations originating in the interior of BC, for example, have greater capacity for physical exertion and can withstand warmer water temperatures compared to salmon that spawn lower down in the Fraser River watershed (Farrell et al. 2008; Eliason et al. 2011). Optimal temperatures for aerobic scope often closely align with historical water temperatures experienced by the population during up-river migration (Lee et al. 2003; Farrell 2007; Farrell et al. 2008; Eliason et al. 2011). Pacific salmon are semelparous, consequently lifetime fitness for an individual depends entirely on a single, successful migration and subsequent spawning event (Mathes et al. 2009; Reed et al. 2011; Eliason and Farrell 2016). The adult life stage, therefore, is critical when considering the overall thermal tolerance of a population.

Fewer studies have looked at aerobic scope in juvenile salmon, a life stage that has very different selective pressures. Juvenile Chinook salmon (*Oncorhynchus tshawytscha*) and coho salmon (*O. kisutch*) from the Horsefly River provide an interesting opportunity to study this topic. A seasonal difference in adult migration timing may result in different T_{opt} between the two species at this life stage, although rearing sympatrically in the same streams as juveniles may suggest a similarity in aerobic scope at this earlier life stage.

The objectives of this study were to determine aerobic scope across a range of experimental temperatures for juvenile Chinook salmon and coho salmon from the Horsefly River and to identify an optimal temperature at which aerobic scope is maximized in each

species. I also wanted to compare results of this study with those from Chapter 2, to investigate whether or not temperature preference aligns with optimal temperature for physiological performance.

METHODS

Fish Husbandry

Chinook salmon and coho salmon for aerobic scope experiments were caught in October 2016, as described in Chapter 1, and held in the Aquatic Animal Holding Facility at the University of Northern British Columbia for the duration of laboratory experiments. These fishes were part of the same group used for temperature preference experiments and fish husbandry methods were identical to methods described in Chapter 2.

Metabolic Rate Measurements

I used a total of 40 Chinook salmon and 14 coho salmon for aerobic scope experiments. Because I caught a limited number of coho salmon in October 2016, the 14 coho salmon used for aerobic scope measurements had also been used for temperature preference experiments. To control for the possible effect of using the same fish for multiple measures, 15 of the Chinook salmon that I used for aerobic scope experiments had previously been tested for temperature preference. To maximize sample size, particularly with the limited number of coho salmon available, I randomly selected eight Chinook salmon and nine coho salmon and repeated aerobic scope experiments with them a second time.

Individuals that underwent multiple experiments had at least five days of rest between aerobic scope trials.

Each fish was only tested at a single temperature, with the exception of the duplicated individuals. I targeted test temperatures of 9, 12, 15, 18, and 21 °C, although actual experimental temperature groupings were 8.8, 13.0, 15.0, 17.4, and 21.0 °C, each ± 0.5 °C. Aerobic scope experiments took place between February 1st and April 7th, 2017.

I measured both routine and maximum oxygen consumption rates in individual, 92 mL respirometer chambers (CH12050, Loligo Systems, Tjele, Denmark), each fitted with 5 mm diameter mini fiber optic oxygen sensor spots (OX11050, Loligo Systems) and submerged in a water bath. Water recirculated between the water bath and the Recirculating Aquaculture System (RAS) to ensure experimental temperature remained constant and that the water bath was continuously flushed with oxygenated water that had passed through the systems' five filtration stages. To circulate water within the chambers I used two peristaltic pumps, which were connected to each chamber by silicone tubing. One pump was continuously on and recirculated water within each chamber to ensure adequate mixing in the closed system so oxygen readings by sensor spots were accurate. The other pump was only switched on during the "flush" phase, pumping fresh water from the water bath through each chamber to replenish dissolved oxygen levels. A four-channel respirometry system (DAQ-PAC-F4, Loligo Systems) continuously measured oxygen concentration in each chamber and temperature in the water bath, allowing for automatic temperature compensation. Data were recorded by AutoResp (v.2.3.0) software (Loligo Systems) every second for the duration of the sampling period.

I selected the first fish of the group of two to four fishes to be tested together randomly. Due to variable oxygen consumption rates by fish of different sizes, it was

important that all fishes in the group were closely matched in size; therefore, I selected subsequent individuals based on visual estimates and then confirmed mass by weighing them. The day before aerobic scope experiments commenced, I transferred fishes from the holding tank to 20-L buckets equipped with air stones with holding temperature water that gradually warmed at a rate of 2 °C/h to the respective experimental target temperature. I monitored water temperature and dissolved oxygen (DO) levels to ensure the appropriate rate of warming and that DO never fell below 90% saturation. At 5 pm, I removed fishes from the buckets and placed them into individual tanks in the second RAS, which was set to the experimental target temperature.

I initiated maximum oxygen consumption measurements at noon each day, where I removed fishes from the second RAS unit, weighed them individually and then placed them into two 20-L buckets filled with water to which they were acclimated. I then manually chased fishes by hand until they no longer reacted to three successive tail pinches within five seconds, which was deemed exhaustion. This occurred within approximately four minutes of chasing. At exhaustion, I transferred fishes into individual respirometry chambers submerged within a water bath thermostatted to the respective temperature and measured maximum oxygen consumption rates. After a sufficient “wait” period in the closed respirometer, oxygen levels fell linearly and I initiated the “measure” phase (lasting from one to four minutes), until DO reached 80% saturation in the first chamber. At this point I triggered the “flush” phase to allow fresh, oxygenated water into the chambers for a recovery period of one to two hours, until fish ventilation slowed and adequate oxygen levels could be maintained in each chamber. I then manually chased fishes to exhaustion a second time and repeated maximum oxygen consumption rate measurements as described above. Following the second measurement, I triggered the “flush” phase for several hours.

After recovery, I initiated the automated intermittent flow respirometry settings to measure routine oxygen consumption until the following morning. I set the “flush” period at 45 minutes for every trial, but adjusted the length of the “wait” and “measure” periods to be appropriate for each group of fishes and experimental water temperature. Larger fish and warmer water temperatures necessitated much shorter measurement periods as they consumed oxygen in the chambers faster than in cases with smaller fish and colder temperatures. Routine oxygen consumption measurements continued until approximately 10 am the following morning. At this time, I removed fishes from the chambers and gradually brought them back down to ambient temperature before returning them to the holding tanks.

Periodically during the course of experiments, I left one or two respirometry chambers empty to determine background oxygen consumption by bacteria present in the water, which was always negligible. I opened the chambers and flushed them with fresh dechlorinated water after each set of experiments. I also sterilized the water bath, chambers and tubing every one to two weeks with Ovadine solution (Syndel, Canada).

Data and Statistical Analysis

Metabolic Rate Determination

AutoResp recorded up to 20 measurements of routine oxygen consumption overnight for each fish. To determine routine metabolic rate (RMR) for each fish, I considered measurements that occurred after 7:00 pm, when the lights turned off and metabolic rate started to approach resting levels, and with R^2 values of 0.90 or greater, as recorded by AutoResp. These values describe goodness-of-fit of linear regressions that the software automatically calculates to determine whether oxygen decrease in the respirometry chamber

was linear over the measurement period. I calculated RMR of the fish as the mean of the lowest three oxygen consumption measurements meeting the above two criteria. I excluded several trials completely when oxygen consumption measurements could not be recorded due to fish mortality following maximum oxygen consumption measurements ($n = 1$), fish recovering but still consuming too much oxygen to set up intermittent flow respirometry for the overnight period ($n = 2$), or equipment malfunction ($n = 5$).

For each fish, I plotted both maximum oxygen consumption traces in Excel and used linear regressions to determine the steepest slope over a 60 second period in either trace with an R^2 value of > 0.93 . In each of these cases, the decline in DO was greater than 5%. Using this slope I calculated maximum metabolic rate (MMR) using the following formula (Steffensen 1989):

$$MMR = \frac{slope \times volume}{mass}$$

where MMR is the maximum rate of oxygen consumption ($\text{mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$), slope is the linear decrease in oxygen over the time period as determined above ($\text{mg O}_2 \text{ L}^{-1} \text{ h}^{-1}$), volume is the effective respirometer volume (L) calculated as the chamber volume minus the volume of the fish (estimated at $1 \text{ kg} = 1 \text{ L}$), and mass is the mass of the fish (kg). I excluded trials from MMR determination when both traces failed to produce a slope that met the above criteria.

For individuals with both RMR and MMR values, I estimated two measures of aerobic scope: absolute aerobic scope ($\text{AAS} = \text{MMR} - \text{RMR}$) and factorial aerobic scope ($\text{FAS} = \text{MMR}/\text{RMR}$). Throughout this thesis, the general term *aerobic scope* refers to AAS, unless otherwise stated. To attribute a temperature to both AAS and FAS, I took the mean of experimental temperatures recorded during RMR and MMR measurements for that individual.

Statistical Analysis

I used R version 3.5.0 (R Core Team 2018) for all statistical analyses and determined statistical significance using an alpha value of 0.05. Means are presented $\pm$ 1 standard deviation (SD). I tested all data for normality using Shapiro-Wilk normality tests and plotted and visually assessed residual plots for normality where appropriate. I tested for homogeneity of variance using Levene's tests if data were normal, or Fligner-Killeen tests if data were not normally distributed.

Within each species I grouped AAS data, and their corresponding RMR and MMR values, into five temperature categories based on the five target temperatures during experiments of 9, 12, 15, 18 and 21 °C. Actual experimental test temperature groupings were 8.8, 13.0, 15.0, 17.4, and 21.0 °C, which included trials conducted at temperatures $\pm$ 0.5 °C. After graphically identifying potential outliers in RMR and MMR data within each temperature group, I tested all suspect points with a Dixon's Q one-tailed test and removed data that were considered outliers at a 95% confidence level. I excluded four Chinook salmon from MMR analyses using temperature groupings because test temperatures were outside of the $\pm$ 0.5 °C range used for grouping. Only one of these four fish had a RMR measurement and subsequent aerobic scope values, therefore, I excluded this single individual in aerobic scope analyses when using temperature groupings.

I independently compared metabolic responses (RMR, MMR, AAS) across temperatures within each species using one-way analysis of variance (ANOVA) tests. If the ANOVA indicated a significant difference, I used a Tukey's Honest Significant Difference (HSD) post hoc test to determine which groups were different from one another.

To investigate if and how aerobic scope varied with experimental temperatures, I evaluated two regression models for each species – linear and quadratic. Regression models

were fitted to individual AAS points for each species, and I chose the best-fitting model as the one with the lowest residual standard error and highest adjusted R^2 value.

To directly compare AAS between species, I performed a two-way factorial ANOVA (type III sums of squares) testing the influence of two categorical independent variables (temperature grouping and species) on AAS as well as the interaction between the two independent variables. Temperature grouping included five levels (9, 12, 15, 18 and 21 °C) and species consisted of two levels (Chinook salmon and coho salmon). I investigated any significant differences using a Tukey's HSD post hoc test.

Finally, because metabolic rate scales with body mass allometrically, it was important to consider fish size both between species and across temperature groupings. I compared mean body mass between Chinook salmon and coho salmon used for aerobic scope determination using a Wilcoxon rank sum test. I also compared body mass within each species across experimental temperature groups using either a one-way ANOVA with normally distributed data, or the non-parametric Kruskal-Wallis test when data were not normally distributed.

RESULTS

In Chinook salmon, RMR, MMR and AAS appeared to be maintained across the range of experimental temperatures (Figure 3.1). There was no identifiable trend in FAS, which ranged from 1.8 to 9.4.

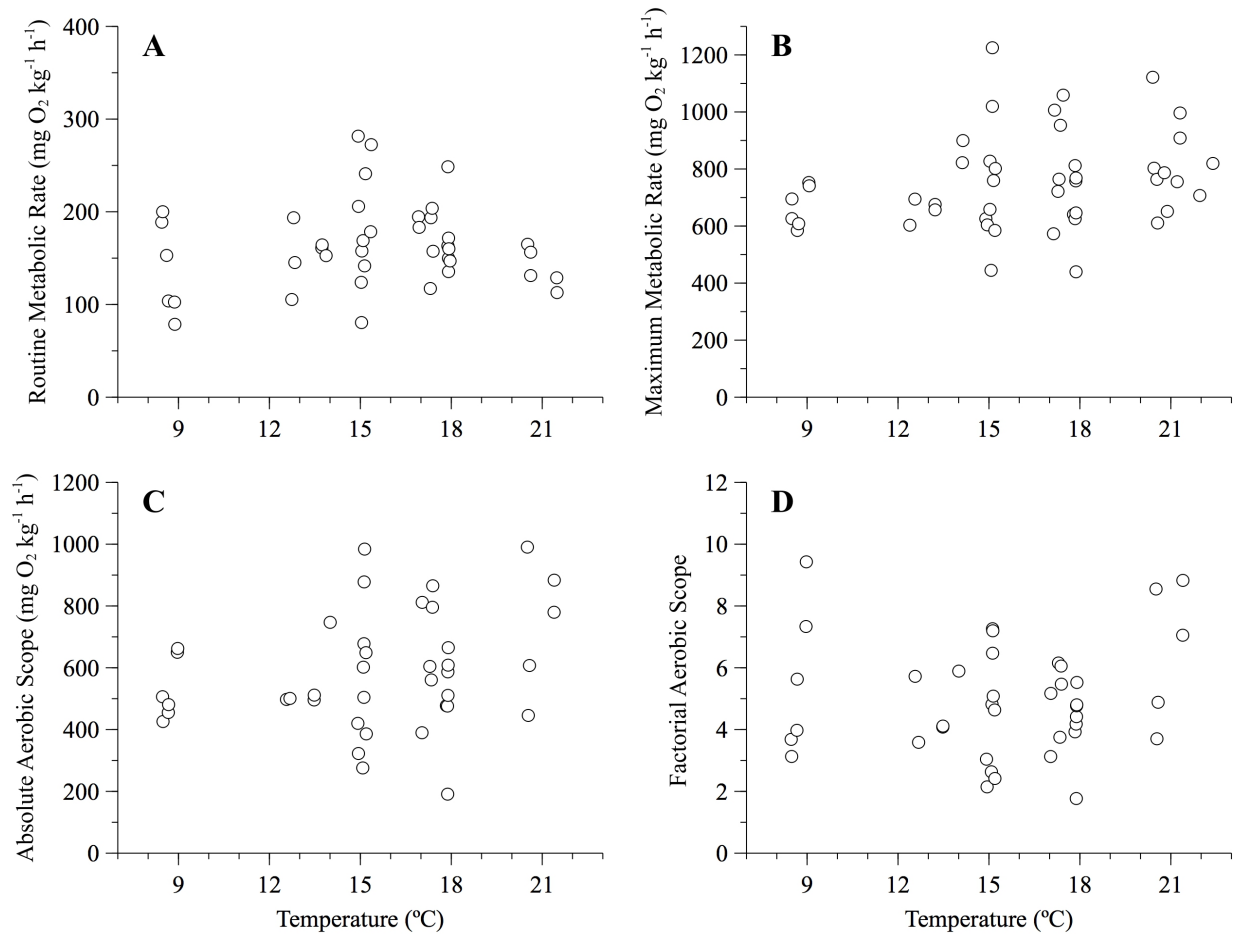


Figure 3.1 Metabolic responses of Chinook salmon to experimental temperatures ranging from approximately 9 °C to 21 °C. Values are (A) routine metabolic rate (n = 40) and (B) maximum metabolic rate (n = 46) measurements and calculated (C) absolute aerobic scope (n = 39) and (D) factorial aerobic scope (n = 39) for individual fish.

After grouping Chinook salmon data into experimental temperature categories, mean MMR and AAS generally increased with experimental temperatures, although the same was not true for RMR. Mean RMR increased from 137.79 ± 50.21 mg O₂ kg⁻¹ h⁻¹ at 9 °C to a high of 185.20 ± 65.08 mg O₂ kg⁻¹ h⁻¹ at 15 °C, but then decreased in the 18 °C and 21 °C groups (Table 3.1). The highest mean MMR (821.94 ± 162.36 mg O₂ kg⁻¹ h⁻¹) and AAS (741.34 ± 217.42 mg O₂ kg⁻¹ h⁻¹) calculated for Chinook salmon both occurred in the 21 °C temperature group. Results of independent one-way ANOVAs testing metabolic responses among

temperature groups in Chinook salmon revealed no significant differences in RMR ($F_{4,34} = 1.56$, $p = 0.21$), MMR ($F_{4,37} = 1.03$, $p = 0.41$), or AAS ($F_{4,33} = 1.24$, $p = 0.32$).

Table 3.1 Mean and standard deviation (SD) of routine metabolic rate (RMR), maximum metabolic rate (MMR) and absolute aerobic scope (AAS) for Chinook salmon within five temperature groups. Mean and SD are presented in $\text{mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$.

	Temperature Groups (°C)				
	9	12	15	18	21
<i>RMR</i>					
n	6	5	10	13	5
mean	137.79	153.90	185.20	171.20	138.85
SD	50.21	32.27	65.08	33.98	21.36
<i>MMR</i>					
n	6	4	10	13	9
mean	667.90	657.49	755.13	751.42	821.94
SD	71.48	39.48	229.13	176.44	162.36
<i>AAS</i>					
n	6	4	10	13	5
mean	530.11	501.42	569.94	580.22	741.34
SD	101.35	7.03	234.18	183.57	217.42

In coho salmon, RMR, MMR and the resulting AAS appeared to increase with experimental temperature (Figure 3.2). There was no identifiable trend in FAS in coho salmon, and it ranged from 2.5 to 10.7 across the range of temperatures tested.

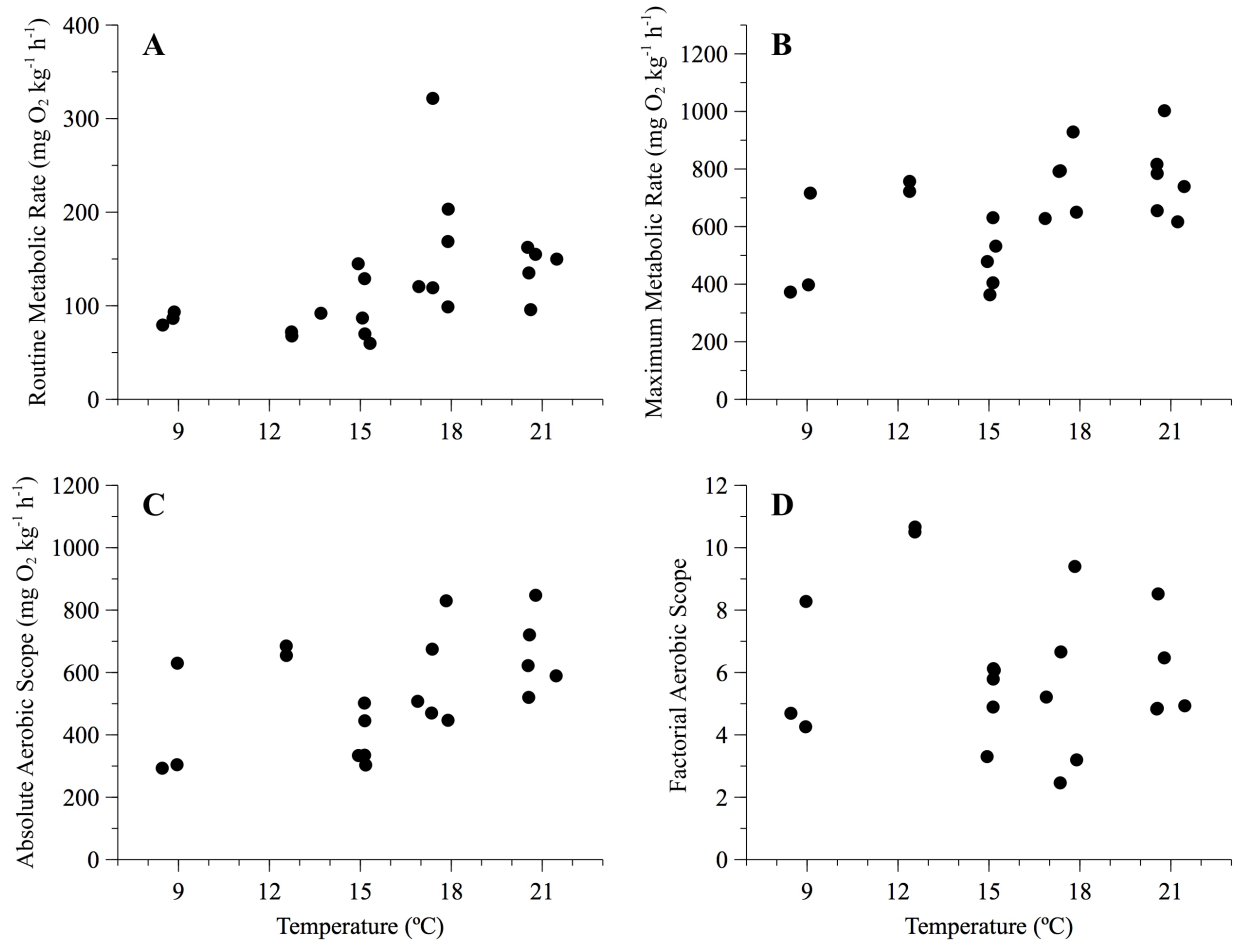


Figure 3.2 Metabolic responses of coho salmon to experimental temperatures ranging from approximately 9 °C to 21 °C. Values are (A) routine metabolic rate ($n = 22$) and (B) maximum metabolic rate ($n = 21$) measurements and calculated (C) absolute aerobic scope ($n = 20$) and (D) factorial aerobic scope ($n = 20$) for individual fish.

The trend of greater RMR, MMR and AAS at higher temperatures largely held true when coho salmon data were combined into temperature groups (Table 3.2). Mean RMR generally increased with warmer temperature, although the highest mean RMR of 172.01 ± 86.67 mg O₂ kg⁻¹ h⁻¹ was in the 18 °C group. This is due to one RMR value in the 18 °C group for an individual coho that was much higher than the rest, but could not be excluded based on outlier criteria. Mean MMR was higher than expected at 12 °C, but was only based on a sample size of two. Because RMR measurements for these two coho salmon were

among the lowest recorded at any temperature, the calculated AAS for the 12 °C group was the highest of any temperature group at $669.71 \pm 21.52 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$. The FAS calculated for these same two coho salmon were the highest in either species at 10.5 and 10.7.

Results of independent one-way ANOVAs testing metabolic responses among temperature groups in coho salmon revealed no significant differences in RMR ($F_{4,17} = 2.89$, $p = 0.054$), but several significant differences in MMR ($F_{4,16} = 5.45$, $p = 0.006$) and AAS ($F_{4,15} = 3.96$, $p = 0.02$) among temperature groups. Mean MMR at both 18 °C ($t = 3.35$, $p = 0.03$) and 21 °C ($t = 3.63$, $p = 0.02$) were significantly higher than mean MMR at 15 °C. Mean AAS at 21 °C was significantly higher than mean AAS at 15 °C ($t = 3.23$, $p = 0.04$).

Table 3.2 Mean and standard deviation (SD) of routine metabolic rate (RMR), maximum metabolic rate (MMR) and absolute aerobic scope (AAS) for coho salmon within five temperature groups. Mean and SD are presented in $\text{mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$. Matching symbols denote significant differences ($p < 0.05$).

	Temperature Groups (°C)				
	9	12	15	18	21
RMR					
n	3	3	5	6	5
mean	86.43	77.27	98.10	172.01	139.64
SD	6.98	12.94	37.18	86.67	26.45
MMR					
n	3	2	5	5	6
mean	495.47	739.61	481.91 ^{*,†}	758.44 [*]	769.04 [†]
SD	191.61	24.58	105.81	122.57	137.15
AAS					
n	3	2	5	5	5
mean	409.04	669.71	383.81 [‡]	585.76	659.89 [‡]
SD	191.19	21.52	85.30	163.05	127.32

There was a 1.4-fold increase in average AAS between 9 °C and 21 °C groups in Chinook salmon, and a 1.6-fold increase in coho salmon (Figure 3.3). The best-fit line for AAS in Chinook salmon was the following non-significant quadratic polynomial

relationship: $y = 1.94x^2 - 42.51x + 755.55$ ($R^2 = 0.098$, $p = 0.16$). The best-fit line for AAS data in coho salmon was the following significant linear relationship: $y = 19.22x + 229.76$ ($R^2 = 0.204$, $p = 0.046$). Results of the two-way factorial ANOVA showed a significant main effect of temperature on AAS ($F_{4,48} = 3.11$, $p = 0.02$). A Tukey HSD post hoc test revealed that significant differences were between the 9 °C and 21 °C groups ($t = 2.84$, $p = 0.048$) and also between the 15 °C and 21 °C groups ($t = 3.12$, $p = 0.02$). Species did not significantly affect AAS ($F_{1,48} = 0.70$, $p = 0.41$), and the interaction between temperature and species was not significant ($F_{4,48} = 1.24$, $p = 0.31$).

Mean body mass of all Chinook salmon (7.25 ± 1.30 g) and coho salmon (7.21 ± 1.54 g) used for aerobic scope determination did not differ significantly between species (Wilcoxon rank sum, $W = 373$, $p = 0.79$). Within Chinook salmon, mass varied significantly among temperature groups (One-way ANOVA, $F_{4,33} = 4.36$, $p < 0.006$). Individual Chinook salmon tested at approximately 9 °C (8.78 ± 1.15 g) were significantly heavier than those at 15 °C (7.05 ± 1.35 g), 18 °C (6.94 ± 1.01 g) and 21 °C (6.44 ± 0.57 g; $p < 0.025$ for each). Within coho salmon, mass did not differ significantly among temperature groups, although it was very close to significant (Kruskal-Wallis, $\chi^2_4 = 9.464$, $p = 0.051$). Coho salmon tested at approximately 9 °C had an average mass of 8.57 ± 1.89 g, while mean mass in other temperature groups ranged from a low of 5.74 ± 0.95 g to a high of 8.06 ± 1.39 g.

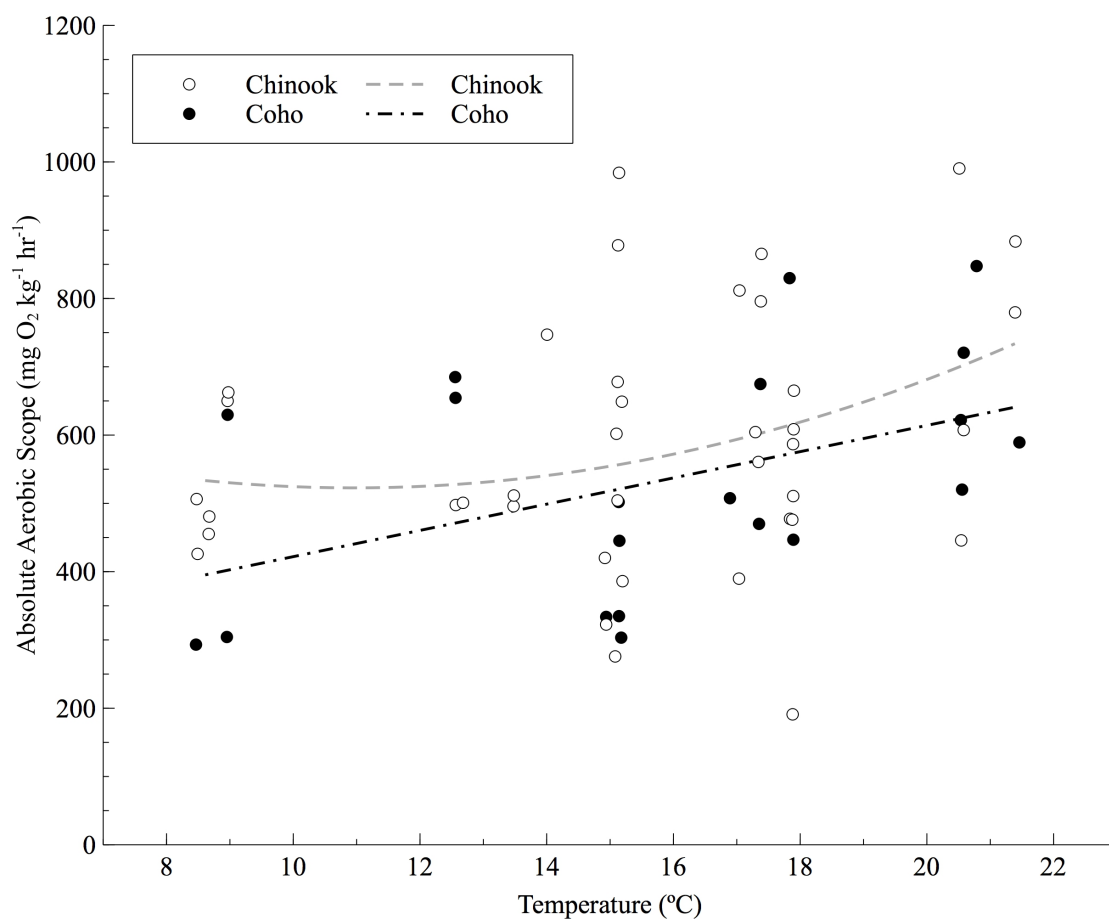


Figure 3.3 Absolute aerobic scope (mg O₂ kg⁻¹ h⁻¹) values for 39 Chinook salmon and 20 coho salmon. Quadratic (Chinook) and linear (coho) regression lines were fit to aerobic scope data, with equations for best-fit lines given in the text.

DISCUSSION

Routine and Maximum Metabolic Rates

The effect of temperature on RMR and MMR was less consistent than expected. In Chinook salmon, RMR and MMR were maintained across the range of experimental temperatures. In coho salmon, RMR and MMR generally increased with temperature. Overall, Chinook salmon had slightly higher values for both RMR and MMR and also appeared to have greater variability in individual measurements compared to coho salmon.

Although seemingly large, this variability was comparable to that presented by Eliason et al. (2013), where up to a three-fold difference in maximum oxygen consumption rates between individuals at a specific temperature were recorded. Even greater variability in both RMR and MMR was reported in a study by Poletto et al. (2017). Intraspecific variation in both RMR and MMR is commonly seen across a range of fish taxa and it is not uncommon to have individuals of the same species, sex and age vary in metabolic rate by two or three fold, even when held under similar laboratory conditions (Norin and Malte 2011; Burton et al. 2011; Killen et al. 2012; Metcalfe et al. 2016; Norin and Clark 2016). Norin and Malte (2011) showed that brown trout (*Salmo trutta*) had high repeatability in successive metabolic rate measurements over the course of several weeks, suggesting that variability in metabolic rate is indicative of true differences in individuals rather than due to inconsistent experimental measurements.

Wide acceptance of Fry's (1947) concept of aerobic scope has promoted the idea that MMR plateaus or begins to decrease at warmer temperatures (Farrell 2009; Pörtner 2010), as it is this trend in MMR that creates the bell-shaped aerobic scope curve (Norin and Clark 2016). Contrary to this pattern, there have been numerous examples, including the present study, of MMR being maintained or continuing to increase across a wide range of experimental temperatures and only plateauing at temperatures close to the upper lethal limit, if at all (reviewed in Norin and Clark 2016).

Aerobic Scope

Absolute Aerobic Scope

Absolute aerobic scope for both Chinook salmon and coho salmon in the current study did not show an obvious peak and AAS was either maintained (Chinook salmon), or continued to increase (coho salmon) across the range of experimental temperatures that spanned approximately 13 °C. The warmest temperature that I tested fish at was 22.4 °C for Chinook salmon and 21.4 °C for coho salmon. It is possible that peaks in AAS exist at warmer temperatures than those tested here; however, I found fishes did not recover well above 22 °C and consequently I did not attempt AS measurements at higher temperatures.

It is also possible that peaks in aerobic scope were obscured by large variability in the data, particularly as a result of variability in MMR measurements. Like intraspecific variation in metabolic rate measurements, large differences in AAS estimates among individuals are also common (Metcalf et al. 2016). Within a single temperature group, both Chinook salmon and coho salmon individuals varied in AAS by factors of up to three, consistent with that seen in adult sockeye salmon (*O. nerka*) (Eliason et al. 2013) and juvenile Chinook salmon (Poletto et al. 2016; Poletto et al. 2017).

There have been some studies done in which aerobic scope closely follows the theoretical “Fry curve” (Fry 1947), increasing with temperature to the optimal temperature (T_{opt}) and then decreasing until eventually reaching an often-fatal critical temperature (Brett 1971; Pörtner and Knust 2007; Farrell et al. 2008; Cooke et al. 2012; Rosewarne et al. 2016). In particular, studies done on adult sockeye salmon in British Columbia (BC) have shown that T_{opt} for aerobic scope is population-specific and closely aligns with historical water temperatures experienced by the population during up-river migration (Lee et al. 2003;

Farrell 2007; Farrell et al. 2008; Eliason et al. 2011; Eliason et al. 2013). In contrast, other studies testing aerobic scope in populations of adult salmon have failed to show that aerobic scope follows the theoretical “Fry curve.” Clark et al. (2011) found that aerobic scope in adult pink salmon (*O. gorbuscha*) continued to increase at temperatures higher than the fish would have experienced at any point in their life cycle, peaking at 21 °C and only starting to decline at approximately 23 °C. Another study done by Raby et al. (2016) testing aerobic scope in adult coho salmon also found that performance was maintained across ecologically relevant temperatures, those that the fish were likely to encounter at any point in their life cycle.

Several studies in juvenile salmonids have yielded similar results to that of Clark et al. (2011) and Raby et al. (2016). In juvenile coho salmon from the Seymour River hatchery in BC, Casselman et al. (2012) found that aerobic scope was maintained or even continued to increase slightly across ecologically relevant temperature ranges. These fish did show a decrease in aerobic scope after 17 °C as temperatures exceeded the ecologically relevant range and approached the upper thermal tolerance limit for juvenile coho salmon from the Seymour River, BC. Juvenile Chinook salmon in California showed no clear peak in aerobic performance, with aerobic scope measurements either remaining stable (Poletto et al. 2017) or increasing linearly (Poletto et al. 2016) across experimental temperatures tested from 12 to 26 °C. Wild juvenile rainbow trout (*O. mykiss*) from the Tuolumne River, California, sustained impressive aerobic capacity over a broad range of temperatures, including maintaining 95% of their peak aerobic scope up to a high of 24.6 °C, which is close to the warmest river temperatures this specific population would experience locally (Verhille et al. 2016).

Factorial Aerobic Scope

Factorial aerobic scope is the proportional increase in metabolic rate above baseline levels to accommodate energetically expensive processes (Clark et al. 2013; Poletto et al. 2016). It is generally accepted that a doubling in metabolic rate (i.e. FAS of 2) is the minimum capacity required to sustain crucial activities such as foraging and digestion (Jobling 1981; Poletto et al. 2016). With the exception of a single Chinook individual (FAS = 1.8), all other Chinook salmon and coho salmon in this study, a total of 58 individuals, had FAS measurements greater than 2. Factorial aerobic scope in both species failed to follow a distinct pattern across the range of experimental temperatures; however, variability in this measure is not uncommon, as even slight differences in RMR (denominator) will have marked effects on the resulting quotients (FAS) (Clark et al. 2013).

While I have included FAS measurements for Chinook salmon and coho salmon in this study, Clark et al. (2013) suggests that AAS often provides a more meaningful and informative measure of aerobic capacity, especially when making comparisons across species. Clark et al. (2013) illustrated this point when they compared a sedentary species to an active species in which there was an eight-fold difference in AAS between the two, but FAS was almost identical. Finally, Clark et al. (2013) also suggested that different activities require a specific amount of oxygen to perform and the proportional increase in metabolic rate above resting levels in some cases is irrelevant.

Fish Size

Both Chinook salmon and coho salmon varied in mean mass across experimental temperature groups either significantly, or close to significantly. In both species, the

individuals tested at approximately 9 °C were heavier than the other temperature groups. Larger fish consume oxygen more quickly than smaller fish; therefore, it was necessary to match individuals in size as closely as possible when testing metabolic rates in groups of up to four fish. While I chose the first fish for a test group randomly, I then selected up to three others that were of approximately the same size. Because of this, the few heaviest individuals were tested together at approximately 9 °C.

Metabolic rate scales with body mass allometrically. This relationship was first described by Kleiber (1932, 1947), and has become known as “Kleiber’s Law” or the “ $\frac{3}{4}$ Rule,” where metabolic rate scales with body size to the power of 0.75. Thus, smaller animals have a higher mass-specific metabolic rate than larger animals. Since then there have been many criticisms of the law, suggesting that a scaling exponent of 0.75 is not ubiquitous and there is in fact no universal mass scaling exponent in regards to metabolism (Post and Lee 1996; Clarke and Johnston 1999; Bokma 2004; Killen et al. 2007; Killen et al. 2010; White 2011; Lucas et al. 2014; Hulbert 2014). Specific to fish, Lucas et al. (2014) determined a scaling exponent of 0.97 for resting metabolic rate in zebrafish (*Danio rerio*) and 0.93 for maximum metabolic rate. Killen et al. (2007) tested metabolic rate in three species of teleost fish across various life-history stages and estimated a scaling exponent for standard metabolic rate of between 0.82 and 0.84, and for maximum metabolic rate of 0.88 to 0.93. Numerous studies across a variety of fish species including salmonids have found similar scaling exponent values ranging from 0.80 to 0.88 (Post and Lee 1996; Clarke and Johnston 1999; Bokma 2004; White et al. 2006).

Oxygen consumption rates, and subsequently RMR, MMR, AAS and FAS, are calculated as mass-specific values that account for fish mass. Across aerobic scope studies, it is standard practice to calculate metabolic rates by dividing oxygen consumption by body

mass to the power of 1. Of the many aerobic scope studies I reviewed, only two used a body mass exponent of 0.95 (Poletto et al. 2016; Verhille et al. 2016). Poletto et al. (2016) justified this by referencing a value halfway between the 0.93 and 0.97 that were estimated by Lucas et al. (2014). In a later paper, the same authors reverted to an exponent of 1 (Poletto et al. 2017). This convention may be due to the fact that the range of body masses used in these studies is relatively small, and so the relationship between body mass and metabolic rate can be assumed to be close to linear. To be consistent with these published studies, I also used a body mass exponent of 1. The 9 °C temperature group included the heaviest individuals of both species; therefore, if anything, metabolic rates for this group would have been a slight underestimation. Perhaps more appropriately, scaling metabolic rates with body mass to the power of some value less than 1 would have flattened the aerobic scope curves to a small degree.

Acclimation Temperature

Acclimation temperature is an important consideration when conducting laboratory studies investigating temperature effects, although it is unclear to what degree it affects aerobic scope measurements and the determination of T_{opt} . Poletto et al. tested two different acclimation regimes in their two studies: 14 and 20 °C (2016) and 15 and 19 °C (2017); in both cases acclimation temperature had no effect on aerobic scope throughout the range of experimental temperatures. In contrast, Verhille et al. (2016) found that a population of rainbow trout in California, near its southern range limit, showed aerobic scope to peak close to local high river temperatures, which was much warmer than peak aerobic scope for rainbow trout from more northern latitudes.

The purpose of this study was primarily to compare aerobic scope between Chinook salmon and coho salmon. I held all fishes under identical laboratory conditions for several weeks prior to experiments commencing; therefore, acclimation temperature in these experiments likely affected both species equally.

Ecological Relevance

Aerobic scope is commonly measured in fishes as a gauge of physiological performance and a means to identify an optimal temperature or temperature range in which it is maximized. Despite its popularity, some researchers have questioned the usefulness and ecological relevance of determining aerobic scope and T_{opt} .

Although generally easy to measure, inconsistent use of a wide range of methods has spurred criticism as results from different studies are difficult to compare (Steffensen 1989; Clark et al. 2013; Norin and Clark 2016; Rosewarne et al. 2016; Svendsen et al. 2016). In addition, varying definitions of standard, routine, resting and maximum metabolic rates themselves have complicated aerobic scope experiments and the consistency across studies (Steffensen 1989; Norin et al. 2014; Norin and Clark 2016; Svendsen et al. 2016).

There are a number of studies that show that aerobic scope peaks at an optimal temperature and then decreases within an ecologically relevant temperature range (e.g. Farrell et al. 2003; Farrell et al. 2008; Lee et al. 2003; Eliason et al. 2011). In contrast, many other studies suggest that aerobic scope often continues to increase through temperature ranges commonly experienced by the fish, only beginning to decrease when approaching an upper lethal temperature, if at all (e.g. Clark et al. 2011; Casselman et al. 2012; Norin et al. 2014; Poletto et al. 2016; Poletto et al. 2017; Raby et al. 2016; Verhille et al. 2016). This

contradiction questions the importance of measuring aerobic scope and the relevance of using this measure in an ecological context.

Another topic of question is, even with a valid laboratory estimate of aerobic scope, how often and in what situations will fishes in the wild actually exploit their aerobic scope capacity? The determination of aerobic scope can become more ecologically important by using other laboratory techniques measuring proxies of metabolic rate and coupling those with studies done in the wild (Norin and Clark 2016). For example, calibrating heart rate (Clark et al. 2011) or accelerometry (Eliason et al. 2013) measures in the laboratory against standard oxygen consumption measurements would then allow researchers to use biotelemetry tags in the wild to measure those same proxies and therefore estimate aerobic scope in real world situations (Metcalf et al. 2016).

While there are significant limitations to using aerobic scope in any meaningful ecological context, the concept still has value in predicting performance with respect to temperature (Farrell 2016). It is important to keep in mind that aerobic scope more accurately indicates a fundamental niche rather than a realized niche as there are many other competing factors that dictate which areas a fish will choose to occupy, such as food quality and availability, competition and predator avoidance (Farrell 2013, 2016). Verberk et al. (2016) recommends a more integrative approach that goes beyond simply using oxygen consumption rates and aerobic scope to predict thermal niches. While the traditional “Fry curve” may identify a T_{opt} for aerobic scope in terms of total capacity, different activities are performed best at specific temperatures that may differ from T_{opt} (Farrell 2016).

In the context of the present study, measuring aerobic scope in Chinook salmon and coho salmon from the Horsefly River adds another valuable interspecies comparison to that of temperature preference (Chapter 2) and upper thermal tolerance (Chapter 4) experiments.

In terms of aerobic scope, these two populations appear to be well adapted for current environmental conditions where they rear sympatrically, rather than show any tendencies to be suited to selective pressures they will experience as adults.

CHAPTER 4

UPPER THERMAL TOLERANCE

ABSTRACT

Upper and lower thermal limits define the range of temperatures that an animal can tolerate.

The objective of this study was to determine the upper limit of temperatures tolerated by

Horsefly River Chinook salmon (*Oncorhynchus tshawytscha*) and coho salmon (*O. kisutch*).

Based on observations during aerobic scope experiments in which Chinook salmon recovered more easily at warmer temperatures than coho salmon did, I expected Chinook salmon to

have a higher upper thermal tolerance than coho salmon. I conducted laboratory experiments following critical thermal methodology, in which I raised water temperature at a steady rate

until loss of equilibrium occurred, which signified critical thermal maxima (CT_{max}). Coho

salmon (27.99 ± 0.25 °C) had a significantly higher CT_{max} than Chinook salmon ($27.73 \pm$

0.33 °C). In coho salmon, CT_{max} did not differ significantly among the three streams fish

were collected from, suggesting stream origin did not affect upper thermal tolerance in this

study. Chinook salmon used for CT_{max} experiments were significantly larger than coho

salmon. There was a significant positive correlation between CT_{max} and fish mass in Chinook

salmon, but no correlation was evident in coho salmon. The ecological relevance of upper

thermal tolerance experiments has been questioned extensively. There are also numerous

ways to measure thermal tolerance, making comparisons among studies complicated. Despite

these limitations, measuring CT_{max} in these two populations was valuable in the context of

temperature preference (Chapter 2) and aerobic scope (Chapter 3) experiments completed

with these same populations of Chinook salmon and coho salmon.

INTRODUCTION

Fishes can tolerate a range of temperatures, bounded by upper and lower thermal limits (Fry 1947; Brett 1956). This thermal zone defines a fundamental niche, although fishes will only occupy smaller ranges within this zone throughout their life history, dictated by factors such as preference (Chapter 2) and physiological performance (Chapter 3).

Two schools of methodology for measuring thermal tolerance developed independently – incipient lethal temperature (ILT) and critical thermal methodology (CTM). First defined by Fry (1947), ILT is the temperature at which 50 percent of the population will not survive under indefinite exposure. Experiments following this methodology (e.g., Hart 1952; Brett 1952) involve abruptly exposing fishes to temperatures above or below acclimation temperature until death occurs (Becker and Genoway 1979; Lutterschmidt and Hutchison 1997; Beitinger et al. 2000). Cowles and Bogert (1944) established CTM through their pioneering ideas of thermal tolerance and how animals react to relatively extreme temperatures with their work on desert reptiles in the Coachella Valley, California. They defined critical thermal maximum (CT_{max}) as “the thermal point at which locomotory activity becomes disorganized and the animal loses its ability to escape from conditions that will promptly lead to its death” (Cowles and Bogert 1944: 277). Refinement of the definition specified CT_{max} as the arithmetic mean of thermal points measured (Lowe and Vance 1955), and further that CT_{max} tests involve heating an animal up at a constant rate from a previous acclimation temperature (Hutchison 1961).

Although both methodologies use time and temperature as experimental variables, final temperatures withstood are not equivalent (Becker and Genoway 1979; Currie et al. 1998; Beitinger et al. 2000). Experiments using ILT methods aim to determine a zone of tolerance, suggested by Fry (1947) as the range of temperatures between lower and upper

incipient lethal levels within which an animal can live indefinitely in regards to temperature effects alone. Methods following CTM reveal temperatures that more accurately represent a zone of resistance, which Fry (1947) defined as a temperature range beyond the zone of tolerance in which animals can survive for a significant length of time, but not indefinitely. Experimental endpoints differ between the methods as well. Using ILT methodology, the endpoint is death, and often 50 percent mortality within a test group is used as the criterion. In contrast, the most common endpoint used in CTM experiments is loss of equilibrium (LOE), although other end points such as muscle spasms are also used (reviewed in Lutterschmidt and Hutchison 1997). In many thermal tolerance experiments, CTM has increasingly become the preferred methodology because it provides a quick measure of acute thermal tolerance using simple methods, requires fewer animals than ILT protocols and has a nonlethal endpoint (Becker and Genoway 1979; Lutterschmidt and Hutchison 1997; Beitinger et al. 2000; Geist et al. 2010).

In this study, I was particularly interested in determining the upper thermal limits for Chinook salmon (*Oncorhynchus tshawytscha*) and coho salmon (*O. kisutch*) from the Horsefly River in central British Columbia. At the upper end of test temperatures (17–21 °C) during aerobic scope experiments (Chapter 3), I observed that Chinook salmon seemed to recover more easily from exhaustive exercise than coho salmon did. With the exception of one Chinook salmon that died in the chamber while recovering from the first maximum metabolic rate (MMR) measurement, all Chinook salmon maintained dorsal-side-up body position in the chamber as they recovered. During some experiments at 17 °C and many at 21 °C, coho salmon lost equilibrium for extended periods of time following one or both of the MMR measurements. Maximum and routine oxygen consumption rates were comparable for the two species throughout the range of temperatures, including at the upper range, although

because of their difficulty recovering, it is likely that coho salmon were closer to their upper thermal tolerance at this point than Chinook salmon to their upper thermal tolerance. Based on these observations, I tested whether there were differences between Chinook salmon and coho salmon in their upper thermal tolerance levels.

METHODS

Fish Husbandry

Chinook salmon and coho salmon for upper thermal tolerance experiments were caught in October 2017, as described in Chapter 1, and held in the Aquatic Animal Holding Facility (AAHF) at the University of Northern British Columbia (UNBC) for the duration of laboratory experiments. The Animal Care and Use Committee at UNBC approved fish holding conditions and experimental procedures (protocol #2017-11). In general, fish husbandry methods were identical to 2016 methods, described in Chapter 2, with the following exceptions. In 2017, I fed fishes exclusively tubifex worms, and there was a three-week holding period from the time I transported fishes to the AAHF to the time I started experiments. As in 2016, water temperature was maintained at approximately 9 °C and I fasted fishes for 24 hours prior to upper thermal tolerance experiments.

Critical Thermal Maxima Measurements

To test acute upper thermal tolerance in both species, I followed CTM originally introduced by Cowles and Bogert (1944), with fish-specific protocols initially developed by Heath (1963, 1967) and Lowe and Heath (1969) and refined more recently by Fangue et al.

(2006; 2011). I used a total of 21 Chinook salmon and 36 coho salmon for $CT_{\max}$ experiments, which I conducted between November 10th and December 1st, 2017.

I tested two fish at a time, which I placed within individual mesh cells submerged in a 35 L water bath. The water bath was equipped with three standard aquarium heaters. I adjusted heater settings throughout the trial as needed to achieve a constant rate of heating. I positioned two air stones in the water bath, one at each end, to maintain high oxygen saturation throughout the trial and to ensure adequate mixing to achieve uniform water temperature. To obtain accurate, real-time water temperature measurements, I secured a temperature probe in the centre of each cell. Probes were connected to temperature analyzer boxes, which communicated with a laptop computer via a data acquisition system (DAQ-S, Loligo Systems). I ran ShuttleSoft (v.2.6.4) software (Loligo Systems) on the laptop computer to monitor water temperatures in each cell in real-time and to record each trial.

To begin the experiment, I placed one fish in each mesh cell with the water bath initially at approximately 9 °C. As the trial progressed, water temperature steadily increased at approximately 0.3 °C/min, a rate widely used in $CT_{\max}$ studies (e.g. Fangue et al. 2006, 2011), and a rate recommended by Becker and Genoway (1979) after completing an extensive literature review of methods in addition to examining their own data. The rate of temperature increase in $CT_{\max}$ experiments is an important balance between being slow enough that a fish's core body temperature closely tracks the water temperature and being fast enough to prevent thermal acclimation to the increasing experimental temperatures (Hutchison 1961; Beitinger et al. 2000). Individual rates of increase were between 0.27 °C/min and 0.33 °C/min for all trials. Water temperature continued to increase until LOE occurred, which I defined as the fish failing to maintain dorsal-side-up body position for at least five seconds. I used LOE as the experimental endpoint at which time I recorded the

water temperature and immediately removed the fish and transferred it to an aerated bucket at approximately 13 °C water temperature. After ten minutes of recovery in the bucket, I weighed each fish and placed it into the holding tanks at an ambient temperature of 9 °C. Recovery following CT_{max} experiments was 100%, with no mortalities during or after experiments.

Between each trial, I drained the water bath completely and re-filled it with fresh dechlorinated water. Every one to two weeks, I sterilized the entire system with Ovadine solution (Syndel, Canada).

Data and Statistical Analysis

All trials were successful, resulting in final sample sizes of 21 Chinook salmon and 36 coho salmon for upper thermal tolerance experiments. I used the LOE temperature as a measure of CT_{max} for individual fishes, and calculated the arithmetic mean within each species to determine a mean CT_{max} for Chinook salmon and coho salmon.

In addition to comparing upper thermal tolerance between species, I also investigated whether or not it differed with stream origin. I compared upper thermal tolerance of coho salmon from Patenuade, Wilmot and Woodjam creeks. All except one Chinook salmon were caught in Patenuade Creek, so a comparison among streams was not possible for this species.

I compared fish mass between the two species as well, as sizes of individual fish used for upper thermal tolerance experiments varied considerably. I also tested for correlations between upper thermal tolerance and mass in each species.

I used R version 3.5.0 (R Core Team 2018) for all statistical analyses and determined statistical significance using an alpha value of 0.05. Means are presented $\pm$ 1 standard

deviation (SD). I used Shapiro-Wilk normality tests for all assessments of data normality. To compare variances between two groups, I used a Fligner-Killeen test of homogeneity of variances and among the three streams I used a Bartlett test of homogeneity of variances. In the comparison of two groups, I used the non-parametric Wilcoxon Rank Sum test when either one or both data sets were not normally distributed, or outliers were present that could not be excluded as erroneous measurements. To compare CT_{max} of coho salmon in Patenaude, Wilmot and Woodjam creeks, I used a one-way analysis of variance (ANOVA) and also plotted and visually observed the residuals. Finally, to test for relationships between fish mass and upper thermal tolerance in each species, I used a Spearman's Rank-Order Correlation.

RESULTS

Critical thermal maxima for 21 Chinook salmon individuals ranged from 26.93 °C to 28.15 °C, with a mean of 27.73 ± 0.33 °C. Coho salmon ($n = 36$) CT_{max} ranged from 27.30 °C to 28.35 °C, with a mean of 27.99 ± 0.25 °C. A Wilcoxon Rank Sum test revealed that CT_{max} for coho salmon was significantly higher than that of Chinook salmon ($W_{55} = 555.5$, $p = 0.003$; Figure 4.1).

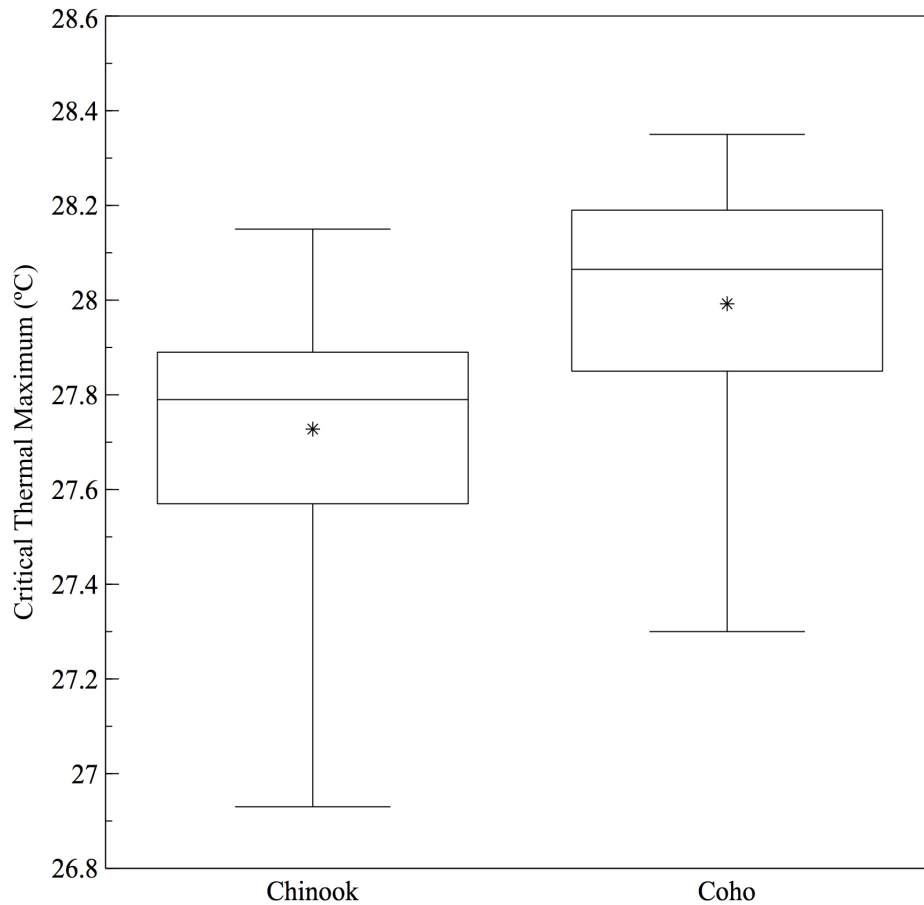


Figure 4.1 Mean critical thermal maxima for coho salmon ($n = 36$) was significantly higher ($p = 0.003$) than for Chinook salmon ($n = 21$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values. Asterisks denote mean values.

The CT_{max} of coho salmon from Patenaude Creek was 27.90 ± 0.33 °C, from Wilmot Creek was 28.02 ± 0.23 °C and from Woodjam Creek was 28.05 ± 0.17 °C. Sample size for each creek was 12. The one-way ANOVA revealed no significant differences among streams ($F_{2,33} = 1.21$, $p = 0.31$; Figure 4.2).

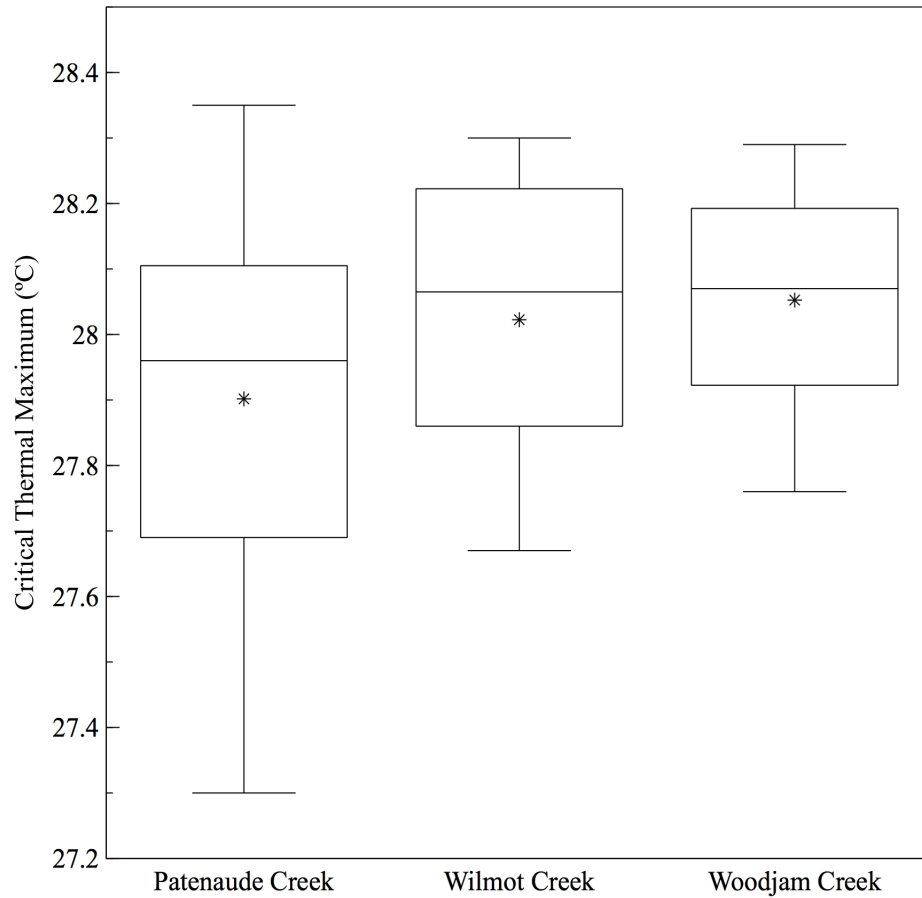


Figure 4.2 Critical thermal maxima of coho salmon ($n = 12$ for each) from Patenaude Creek, Wilmot Creek and Woodjam Creek did not differ significantly ($p = 0.31$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values. Asterisks denote mean values.

The 21 Chinook salmon used for upper thermal tolerance experiments weighed an average of 5.87 ± 0.91 g, while coho salmon ($n = 36$) had a mean mass of 5.09 ± 1.11 g at the time of experiments. Results of a Wilcoxon Rank Sum test indicated that Chinook salmon weighed significantly more than coho salmon ($W_{55} = 560$, $p = 0.003$; Figure 4.3). Spearman's Rank-Order correlations determined that there was a medium-sized (Cohen's effect size), positive correlation between upper thermal tolerance and mass in Chinook salmon ($r_s(19) = 0.49$, $p = 0.02$), but no correlation in coho salmon ($r_s(34) = 0.14$, $p = 0.41$).

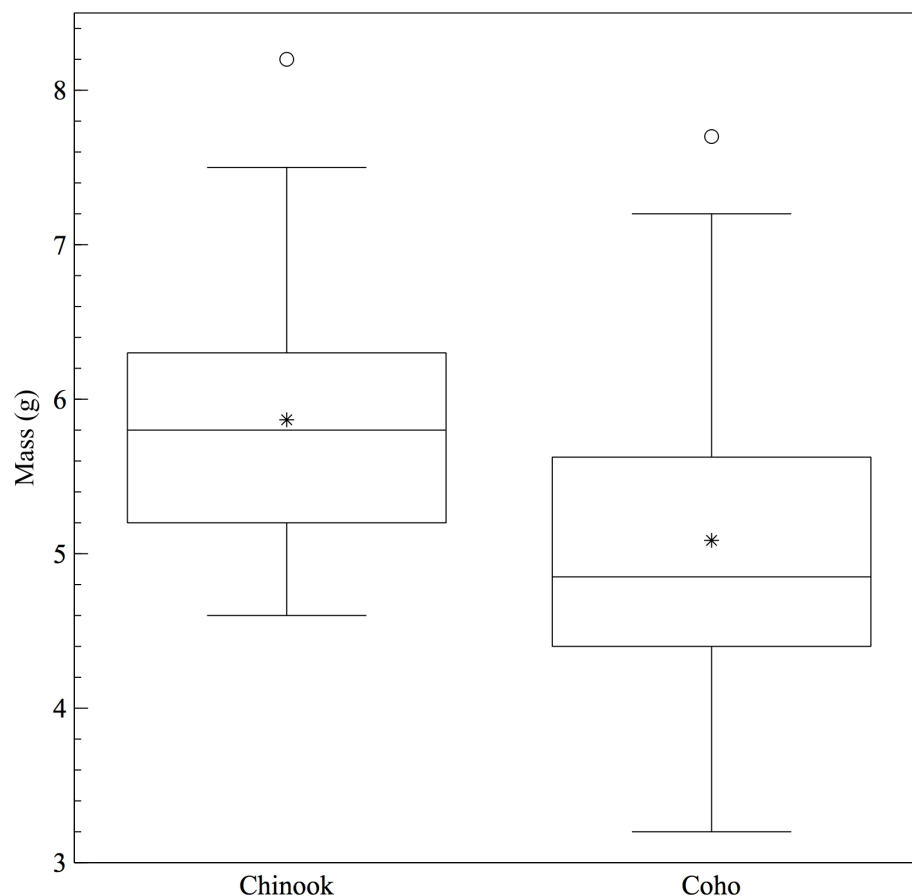


Figure 4.3 Mean mass of Chinook salmon ($n = 21$) used for upper thermal tolerance experiments was significantly heavier ($p = 0.003$) than coho salmon ($n = 36$). Boxes show interquartile ranges, with first quartile, median and third quartile. Whiskers denote minimum and maximum values, with outliers represented by open circles. Asterisks denote mean values.

DISCUSSION

Critical Thermal Maximum

Results from upper thermal tolerance experiments suggested coho salmon (mean $CT_{\max} = 27.99\text{ }^{\circ}\text{C}$) could tolerate slightly, but significantly, higher temperatures than Chinook salmon (mean $CT_{\max} = 27.73\text{ }^{\circ}\text{C}$), which did not support my original observations. Data from both Chinook salmon and coho salmon had low variance and spread; therefore, even a slight

difference between species resulted in statistical significance. Coho salmon mean CT_{max} was only 0.26 °C higher than that of Chinook salmon, which is likely not biologically meaningful. Streams are heterogeneous, consisting of pools and riffles, warm and cold water refugia, varying degrees of cover and large woody debris among countless other characteristics (Beschta et al. 1987; Quinn 2005; Richter and Kolmes 2005). While rearing in these habitats, juvenile salmon experience changes in temperature throughout the day that are much larger than 0.26 °C. The tightness of CT_{max} values (Chinook – SD = 0.33 °C; coho – SD = 0.25 °C) was expected in these experiments, as one known advantage to using LOE as a CTM endpoint is that standard deviations are often less than 1 °C (Beitinger et al. 2000).

Critical thermal maxima determined in my experiments were comparable to other studies done on upper thermal tolerance of Chinook salmon and coho salmon using CTM methods. Muñoz et al. (2014) tested CT_{max} in 25 unique families produced from a full-factorial breeding design of Chinook salmon from Big Qualicum River, BC, origin; mean CT_{max} for all families was 26.5 ± 1.0 °C. Chen et al. (2015) tested CT_{max} in a wild-type population of coho from Chehalis River (BC) origin. Mean CT_{max} for the 18 fish tested was 27.0 ± 0.01 (mean $\pm$ SE). Becker and Genoway (1979) tested CT_{max} in juvenile coho salmon using a variety of heating rates and acclimation temperatures. At the same heating rate as my experiments (0.3 °C/min), mean CT_{max} at 5 °C acclimation was 25.32 °C, and at 15 °C acclimation was 28.70 °C (Becker and Genoway 1979). In my study, fishes were held at approximately 9 °C prior to experiments, therefore my CT_{max} results are consistent with this trend and fall between values Becker and Genoway (1979) found for 5 °C and 15 °C acclimation groups.

While not directly comparable to my results due to differences in methods, earlier studies following ILT methods found lower values for upper thermal tolerances of Chinook

salmon and coho salmon. Using 50% mortality as an endpoint, and also testing prolonged exposure to high temperatures, Brett (1952) found upper lethal temperatures for Chinook salmon (25.1 °C) and coho salmon (25.0 °C) that were several degrees cooler than my results. In studies reviewed by Beschta et al. (1987) using ILT methods, none of the juvenile salmon species could tolerate temperatures over 25.5 °C, even at the warmest acclimation temperatures.

Stream Origin and Acclimation Temperature

Juvenile Chinook salmon and coho salmon are highly mobile while rearing in the Horsefly River watershed (R.L. Case & Associates 2000; Shrimpton et al. 2014; BC MoFLNRORD 2018). Individuals of each species that I caught in the three different streams, therefore, are genetically similar, although they likely experienced different environmental conditions that could affect their upper thermal tolerance. To investigate any differences in upper thermal tolerance among streams, coho salmon from each of the three creeks provided large enough sample sizes for the comparison ($n = 12$ for each). Coho from these three creeks did not differ significantly in CT_{max} ; therefore, similar to results seen with temperature preference (Chapter 2), stream origin does not appear to have an effect on upper thermal tolerance for coho salmon. Mean daily temperatures in Patenaude, Wilmot and Woodjam creeks over a two-week period prior to fish capture in October 2017 averaged between 3.8 °C and 4.9 °C. The three streams varied more considerably when looking at the two-month period prior to capture. From August to October 2017, the coolest of the three streams was Patenaude Creek (average mean daily temperature = 7.1 °C), followed by Wilmot Creek (average mean daily temperature = 8.4 °C) and then Woodjam Creek (average mean daily

temperature = 9.1 °C). Despite these differences in stream temperatures, all fish were held for three weeks in the AAHF under identical conditions before experiments commenced, therefore recent thermal history was the same.

Konecki et al. (1995a) tested CT_{max} in three different populations of coho salmon fry immediately after their capture from creeks in western Washington, and then two of the populations again after they were acclimated to laboratory conditions. When tested in the field immediately following capture, these populations showed significant differences in CT_{max} . Coho salmon from the cooler stream had a lower CT_{max} than the two populations from relatively warmer streams. Laboratory tests with two of the populations resulted in mean CT_{max} of 27.6 and 27.9 °C for fish acclimated to 11 ± 1 °C, which was not a significant difference (Konecki et al. 1995a). This study suggests that differences in CT_{max} among populations were a result of the acclimation temperature and recent thermal history rather than genetic differences.

Acclimation temperature has repeatedly been shown to affect thermal tolerance determined in laboratory experiments (Brett 1956; Becker and Genoway 1979; Jobling 1981; Beschta et al. 1987; Currie et al. 1998; Beitinger and Bennett 2000), and may be the most influential environmental factor to do so (Lutterschmidt and Hutchison 1997). A commonly observed phenomenon is that fishes acclimated to warmer temperatures (e.g. 20–24 °C) can tolerate warmer temperatures and for longer periods than cooler acclimated fishes up until approximately 17 hours, after which point mortality rates coincide with those of fishes acclimated to 15 °C (Brett 1952).

Fish Size

Investigating the possible effect of fish size on CT_{max} in these experiments revealed interesting results. The Chinook salmon I used to test upper thermal tolerance were significantly heavier than the coho salmon. While there was no apparent correlation between mass of fish and CT_{max} in coho salmon, there was a positive correlation in Chinook salmon, which suggests that heavier Chinook salmon tolerated warmer temperatures than smaller individuals. These two results might infer that Chinook salmon would have a higher CT_{max} than coho salmon, although the opposite was true. Because of this, the difference in mass between species likely did not affect the overall outcome of coho salmon tolerating warmer temperatures than Chinook salmon. If anything, the disparity in mass would have understated the difference in CT_{max} between species.

Brett (1952) found that there was no effect of size on high temperature tolerance. Becker and Genoway (1979) suggested that size may affect CT_{max} , but only at rapid rates of increase such as 1 °C/min, when larger fishes may experience a delay in internal tissue warming due to larger body size. They believed this would not be the case in slower heating rates, where internal body temperatures of both small and large animals should track water temperature in a similar manner (Becker and Genoway 1979).

Ecological Relevance

The ecological relevance of experimentally determining upper thermal tolerance has been questioned since the 1950s when it first became a commonplace measure in fishes. A common argument is that in most cases of upper thermal tolerance experiments, fishes can withstand temperatures that greatly exceed the highest environmental temperatures likely to

be seen in natural settings (Hart 1952; Brett 1956; Magnuson et al. 1979). Beschta et al. (1987) suggested that during freshwater rearing stages in particular is when salmonids can tolerate extreme high temperatures, many degrees warmer than they would likely experience during this life stage. This was true in the study done by Konecki et al. (1995a), as both field and laboratory CT_{max} measurements (ranging from 27.6 °C to 29.23 °C) exceeded most maximum summer temperatures for streams in western Washington, where the study populations originated. In response to these criticisms, others have suggested that it is the extreme conditions that exert more selective pressure, even in their rare occurrences, rather than mean temperatures or typical conditions (Huey and Kingsolver 1989; Lutterschmidt and Hutchison 1997). Lowe and Heath (1969) determined CT_{max} for desert pupfish (*Cyprinodon macularius*) in the field at various times of year as well as in laboratory experiments and found that thermal tolerance was highest in the field in June at 44.6 °C, which was only slightly above the summer high stream temperature measured of 41 °C. The difference is likely even smaller, as the authors took this stream temperature measurement in June of a single summer, not during the hottest summer month nor the hottest summer during the study, so the maximum habitat temperature is presumably higher than 41 °C (Lowe and Heath 1969). Winter CT_{max} measurements were only slightly lower than in summer at 36.7 °C in December, much higher than maximum winter stream temperatures of approximately 20 °C (Lowe and Heath 1969). Interestingly, winter CT_{max} is lower than summer stream temperatures, therefore, the desert pupfish must make physiological adjustments seasonally to compensate for environmental temperatures (Lowe and Heath 1969).

In a second argument, Brett (1956) suggested that activities crucial to continued survival of fishes such as feeding, reproduction, competition, predation avoidance and disease resistance likely cease at much lower temperatures than upper lethal limits; therefore

the practical thermal limit is much lower than that seen in an experimental setting. More recent studies have investigated how organ systems central to supporting the above-mentioned activities begin to decline before temperatures reach CT_{max} (e.g. Casselman et al. 2012; Anttila et al. 2013; Muñoz et al. 2014). The oxygen- and capacity-limited thermal tolerance hypothesis is one explanation of how temperature constrains the persistence of organisms in their natural habitats, stating that a functional thermal tolerance range is limited by loss of aerobic scope (Pörtner 2002; Pörtner and Knust 2007; Pörtner and Farrell 2008; Muñoz et al. 2014). As temperatures increase above T_{opt} , heart rate will reach a maximum level and the cardiorespiratory system will fail to deliver the required oxygen to tissues (Pörtner and Farrell 2008; Eliason et al. 2011; Eliason et al. 2013; Anttila et al. 2013; Farrell 2013; Muñoz et al. 2014). At both high and low temperatures where aerobic scope is minimal, there is little ability to support activity above routine needs, thereby impeding critical life functions and reducing survival of the organism (Pörtner and Knust 2007; Farrell et al. 2008; Farrell 2009). Muñoz et al. (2014) linked these ideas directly in their study on juvenile Chinook salmon, where they found that arrhythmic heartbeat and maximum heart rate occurred at temperatures 4.1 °C and 5.3 °C lower than CT_{max} , respectively. Similarly, Chen et al. (2015) found arrhythmic heartbeat to occur several degrees lower than CT_{max} in juvenile coho salmon.

In addition to questioning whether or not the idea of upper thermal tolerance is ecologically relevant, it is also important to consider the validity of the measure itself. Becker and Genoway (1979) suggested that CT_{max} may not be an accurate measure of absolute upper thermal tolerance, but instead a reflection of the experimental methods used. They found that higher acclimation temperatures and a faster rate of warming both elicited higher CT_{max} temperatures under experimental conditions (Becker and Genoway 1979). In addition to

acclimation temperature and rate of warming, the endpoint used also heavily influences CT_{max} , and only experiments with commonality in all three methods should be directly compared (Beitinger et al. 2000). Despite some standardization, results among studies remain hard to compare due to differences in heating rates, thermal history and defined endpoints (Lutterschmidt and Hutchison 1997).

With these limitations in mind, the purpose of measuring upper thermal tolerance in my study was not necessarily to determine a functional upper thermal limit for these populations of Chinook salmon and coho salmon, but rather to supplement findings of aerobic scope experiments (Chapter 3). Both species struggled to recover from MMR measurements at approximately 21 °C; therefore, I did not attempt warmer test temperatures. Aerobic scope was maintained across experimental temperatures in both Chinook salmon and coho salmon. Potentially at warmer temperatures closer to CT_{max} , aerobic scope might have decreased as temperatures approached CT_{max} and T_{crit} could have been defined.

EPILOGUE

In this thesis, I compared temperature responses of two populations of wild salmon from the Horsefly River watershed in central British Columbia (BC). Chinook salmon (*Oncorhynchus tshawytscha*) and coho salmon (*O. kisutch*) from this area reside sympatrically as juveniles during the freshwater rearing stage, but migrate back as adult spawners at different times of the year and therefore experience drastically different water temperature and flow levels at this life stage. Because of these differences in life-history strategies, juveniles of these two populations provided an interesting opportunity to investigate environmental adaptations, specifically to water temperature, in terms of thermal preference, performance and tolerance. This study provided insight into whether these fishes are adapted to their current, shared environment as juveniles, or if they exhibit differences that may be adaptive later in their lives as adults.

I conducted three laboratory studies with juvenile wild-caught Chinook salmon and coho salmon from the Horsefly River to determine temperature preference (Chapter 2), aerobic scope as a function of temperature (Chapter 3) and upper thermal tolerance (Chapter 4), and to compare these measures between species. Temperature preference did not differ significantly between Chinook salmon and coho salmon. Aerobic scope (difference between maximum (MMR) and routine metabolic rates (RMR)) was also similar between the two species, either being maintained (Chinook salmon) or continuing to increase gradually (coho salmon) with experimental temperature. Neither species showed a distinct optimum temperature for peak physiological performance. In upper thermal tolerance experiments, coho salmon could withstand significantly warmer temperatures at their upper thermal limit

compared to Chinook salmon, although the difference was only 0.26 °C, which is likely not biologically meaningful.

It is commonly thought that fishes will behaviourally choose temperatures close to those that will optimize physiological performance (Brett 1971; Magnuson et al. 1979; Jobling 1981; Coutant 1987). In this study, however, aerobic scope continued to increase with experimental temperature up to the maximum temperature tested of approximately 21 °C. Both Chinook salmon (11.73 ± 1.70 °C) and coho salmon (12.49 ± 1.24 °C) preferred temperatures much lower than this, which appear to align more closely with temperatures for optimal growth (Brett et al. 1969; Reiser and Bjornn 1979; Beschta et al. 1987; Bell 1990; Richter and Kolmes 2005). These results seem logical, as the priority for salmon during the juvenile life stage is to maximize growth (Brett et al. 1969; Brett 1971). Unlike adult salmon returning upriver to spawn, which have ceased feeding and expend nearly all energy reserves on the physiologically demanding migration journey, juvenile salmon rearing in a stream do not regularly need to perform at their peak physiological capacity.

Upper thermal tolerance experiments revealed that both Chinook salmon (27.73 ± 0.33 °C) and coho salmon (27.99 ± 0.25 °C) could tolerate temperatures much warmer than the experimental temperatures at which I measured aerobic scope. Unfortunately, due to impaired recovery from MMR measurements at approximately 21 °C, I did not attempt experimental temperatures much warmer than this. I would predict, however, that somewhere between 21 °C and the critical thermal maximum for each species, aerobic scope would peak and then decrease rapidly to a critical temperature where aerobic capacity is zero.

The results of the three experiments show that there is little to no interspecific differences in temperature preference, aerobic scope and upper thermal tolerance in these populations at the juvenile life stage. Overall, these two populations appear to be similarly

adapted for their common environmental conditions as juveniles rather than show any tendencies to be differentially suited to selective pressures they will experience as adults. This research will ultimately contribute to a greater understanding of physiological responses in salmon to different temperature regimes, as well as how physiological requirements relate to behavioural temperature preference and upper thermal limits.

Stream temperature data and results from laboratory experiments in the current studies can build upon previous research done in the watershed with these populations of fishes to help inform specific management decisions for Horsefly River Chinook salmon and coho salmon. In this watershed, both species are known to be highly mobile, moving into several different rearing habitats during the freshwater life stage (Shrimpton et al. 2014; Warren 2009). Warren (2009) found that coho salmon in this area used smaller tributaries to rear, and those which were not necessarily their natal streams. In fact, coho salmon were found to emigrate out of McKinley Creek where they were spawned and into smaller, cooler tributaries prior to peak summer temperatures (Warren 2009). Although my catch effort was not equal among locations or seasons, I successfully caught fishes in the three tributaries that remained the coolest in summer, which appears to corroborate previous findings by Warren (2009). Although both species could perform well at the upper end of temperatures they would normally experience in the wild and could tolerate temperatures several degrees warmer than this, preferred temperatures for both Chinook salmon and coho salmon were considerably cooler at approximately 12 °C. The recovery plan for interior Fraser coho salmon points to habitat destruction as one of the predominant threats to the population and suggests protection of current habitat and rehabilitation of impacted areas as key strategies to help increase fish numbers (Interior Fraser Coho Recovery Team 2006). The recovery plan goes on to identify key rearing habitat as natal streams where spawning occurs (Interior

Fraser Coho Recovery Team 2006), which would be the mainstem Horsefly River and large tributary McKinley Creek within the study watershed. Based on findings in the present studies as well as previous observations of juvenile salmon in the watershed (e.g. R.L. Case & Associates 2000; Warren 2009; Shrimpton et al. 2014; BC MoFLNRORD 2018), I would suggest that smaller tributaries need to be more heavily considered in the management approach.

The Horsefly River watershed is a heavily altered system, largely due to agriculture, ranching and forestry, industries that have been operating in the Horsefly River watershed since the late 1880s (R.L. Case & Associates 2000; BC MoE 2006; Holmes 2008, 2009). Land use has caused destruction of riparian zones adjacent to the Horsefly River mainstem and many of its tributaries, subjecting the system to extreme daily and seasonal temperature fluctuations because of a reduction in shade vegetation (R.L. Case & Associates 2000; BC MoE 2006). Summer water temperatures in the Horsefly River system, which approach 23 °C in some areas, are likely stressful for juvenile Chinook salmon and coho salmon that rear there. Stream and riparian restoration initiatives need to be a primary consideration in any management plan aimed to protect these fish populations in the Horsefly River watershed. While not implying causation, the three streams where I successfully caught both Chinook salmon and coho salmon had all undergone extensive restoration work in the last ten years. In addition to moderating water temperatures, a healthy riparian area promotes stream heterogeneity, supports large woody debris important for cover and availability of insects and other prey, and provides bank stabilization that is critical for maintaining stream integrity and limiting nutrient loading and sedimentation (Beschta et al. 1987; Meehan 1991; Konecki et al. 1995). While working in the watershed, I noticed many instances of cattle entering

streams; therefore, educating landowners about local fish populations and encouraging the use of exclusion fencing would greatly support this cause.

Response to temperature in fishes is an increasingly important topic of study in the face of global climate change. Climate change models suggest that water temperatures will continue to increase and total freshwater discharge will decrease, with seasonal freshet patterns being shifted temporally (Morrison et al. 2002; Crozier and Zabel 2006; Pörtner and Farrell 2008; Reed et al. 2011). The Fraser River is an area of primary focus in BC as it is the largest producer of Chinook salmon (DFO 1999) and sockeye salmon (*O. nerka*) (DFO 2003) in Canada and also supports threatened populations of interior Fraser coho salmon (DFO 2002; COSEWIC 2016) among many other anadromous and resident fishes. According to records dating back to the 1940s, peak summer temperatures in the Fraser River have increased at a rate of approximately 0.2 °C per decade and global climate change models predict that this trajectory will continue into the future (Morrison et al. 2002; Ferrari et al. 2007).

An area for future research could be to test similar temperature responses in these same populations of salmon during the adult life stage. Most adult salmon complete the physically demanding migration during peak summer temperatures and have limited resistance to warming water temperatures projected by climate change models (Sauter et al. 2001; Farrell 2008; Pörtner and Farrell 2008; Reed et al. 2011; Cooke et al. 2012). As semelparous species, this single migration and spawning event for an individual is under strong selective pressure as it determines lifetime fitness for Pacific salmon (Mathes et al. 2009; Reed et al. 2011; Eliason and Farrell 2016). River temperature is critically important to migration success as warm water temperature is strongly correlated with pre-spawn mortality (Gilhousen 1990; Rand and Hinch 1998; Morrison et al. 2002; Ferrari et al. 2007; Farrell et

al. 2008; Mathes et al. 2009). Water temperatures as low as 20 °C, which the Fraser River has surpassed on occasion (Morrison et al. 2002), can negatively affect spawning success (Gilhousen 1990).

Thermal tolerance varies widely among different populations of salmon; therefore, warming river temperatures are expected to elicit population-specific responses (Farrell et al. 2008; Eliason et al. 2011; Clark et al. 2011; Cooke et al. 2012). For example, Clark et al. (2011) showed that pink salmon (*O. gorbuscha*) had a T_{opt} for aerobic scope of 21 °C, higher than any other salmon species tested, which may suggest that pink salmon have a selective advantage and will persist more readily in warmer temperatures compared to other Pacific salmon. Population-specific thermal tolerance ranges are frequently used to predict migration success for a given stock of fish, thereby linking knowledge of physiological scope to biological fitness in the wild population (Cooke et al. 2012; Eliason and Farrell 2016). It is crucial to understand how different populations and life stages of salmon will react to warming temperatures as many species are quickly approaching the upper end of their viable thermal ranges (Crozier and Zabel 2006; Farrell 2008; Pörtner and Peck 2010; Eliason and Farrell 2016).

Differences in life-history strategies among species of Pacific salmon, but sympatric distribution during specific life stages, make species of the genus *Oncorhynchus* ideal study organisms to investigate environmental responses and adaptations. It is likely that results based on populations of Chinook salmon and coho salmon from the Horsefly River watershed in this study could be extrapolated to other interior populations of salmonids in BC and potentially a wider range of fishes. More broadly, salmon are highly sensitive to changes in the environment and therefore act as biological indicators – observations of these species

can be used to draw conclusions about ecosystem changes on larger temporal and spatial scales (Hyatt and Godbout 1999; Reed et al. 2011).

Pacific salmon exist in almost every freshwater river and lake system in the province and are considered keystone species in BC (Hyatt and Godbout 2000). Salmon are an integral part of the ecosystem, acting as both apex predators as well as prey, and supplying critical marine-derived nutrients to aquatic and riparian systems after they spawn and die (Hyatt and Godbout 2000). Salmon hold great intrinsic and ecological importance, but are also an economically valuable resource in the province. For many First Nations communities in BC, salmon are a predominant traditional food source and are central to social and ceremonial customs. I hope that findings from these studies will contribute knowledge and be built upon to help improve conservation and management practices of salmon in BC.

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